

FUNCTIONS



SYNOPSIS

➤ Function or mapping :

A relation 'f' from a set A to a set B is said to be function or mapping if every element of set A has associated with unique element in set B. It is denoted by $f : A \rightarrow B$.

- **Image and Pre-Image:** If 'f' is a function from A to B and $(a, b) \in f$, then $f(a) = b$, where 'b' is called the image of 'a' under f and 'a' is called the pre-image of 'b' under 'f'.

Note: The number of functions from A to B is $\{n(B)\}^{n(A)}$

- **Domain, codomain, Range :** Let $f : A \rightarrow B$ be a function then A is called the domain and B is called the co-domain of the function f.

- If $f : A \rightarrow B$ is a function, then $f(A)$, the set of all f-images of elements in A, is called the range of f

- The range of a function f denoted by $f(A)$, and $f(A) \subseteq B$.

➤ One-one function (Injection) :

A function $f : A \rightarrow B$ is one-one (injection) if distinct elements of A have distinct images in B.

Let $a_1, a_2 \in A$ and $a_1 \neq a_2 \Rightarrow f(a_1) \neq f(a_2)$

(or) $a_1, a_2 \in A$ and $f(a_1) = f(a_2) \Rightarrow a_1 = a_2$ only.

- Let $n(A) = r$ and $n(B) = n$ then the condition to define an injection from A to B is $r \leq n$ and the number of such injections is ${}^n P_r$.

- **Note:** If $r > n$ then the number of injections is 0.

- **Many-one function:** (i) Let $x_1, x_2 \in \text{domain}$ of f. If $x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$ for every x_1, x_2 in the domain, then f is one-one, other wise many-one. (ii) Consider any two points $x_1, x_2 \in X$. Take $f(x_1) = f(x_2)$ and solve the equation if we get $x_1 = x_2$ only then f is one-one other wise it is many-one.

(iii) **Horizontal line test :** If any straight line parallel to x-axis intersects the graph of the function at most at one point then the function is one-one, other wise it is many-one.

(i.e., intersects the graph of the function in atleast two points).

(iv) All even functions are many-one

(v) If the function is continuous and entirely strictly increasing or strictly decreasing in the domain, then f is one - one, other wise many-one.

(vi) All periodic functions are many one.

➤ Onto function (Surjection) :

$f : A \rightarrow B$ is called an onto or a surjection if every element of B has atleast one pre-image in A.

- If $f : A \rightarrow B$ is onto (a surjection) then the range = codomain of 'f' i.e., $f(A) = B$.

- The condition for a function $f : A \rightarrow B$ to be a surjection from A to B is $n(A) \geq n(B)$.

- If $n(A) = n(\geq 2)$, $n(B) = 2$ then the number of onto functions from A to B is $2^n - 2$

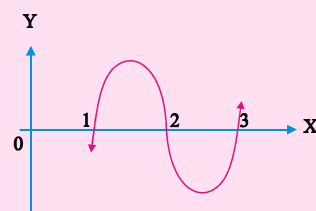
- If $n(A) = r$ and $n(B) = n(\geq 2)$. Then the number of surjections from A to B is $n^r - {}^n C_1(n-1)^r + {}^n C_2(n-2)^r - {}^n C_3(n-3)^r + \dots + (-1)^{n-1} \cdot {}^n C_{n-1}$.

- **Note:** If $n(A) < n(B)$ then the no. of onto functions is zero.

W.E.1: $f : R \rightarrow R$ defined by

$f(x) = (x-1)(x-2)(x-3)$ then $f(x)$ is

Sol:



Graphically, $y = f(x) = (x-1)(x-2)(x-3)$, which is clearly many - one and onto.

- **Into function :** If $f : A \rightarrow B$ is not onto then it is called an into function. i.e., there is no pre-image for atleast one element of B in A then f is into function.

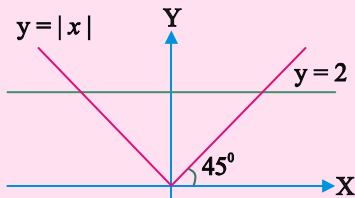
➤ **Working Rule for checking whether the function $f : A \rightarrow B$ is onto or into:**

(i) Let $y \in B$ and $y = f(x)$ and from this find x in terms of y . $\forall y \in B$ if there exists atleast one $x \in A$, then f is onto. Otherwise into. i.e., if range of $f = B$, then f is onto, other wise it is into.

(ii) **Horizontal line test :** If every straight line parallel to x-axis from points in the codomain intersects the graph of the function at atleast one point then f is onto. If the line does not cut the graph of $y = f(x)$ then f is into.

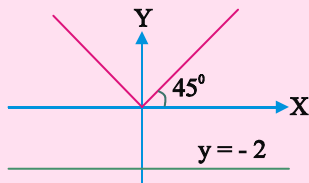
W.E.2: $f : \mathbb{R} \rightarrow [0, \infty)$, $f(x) = |x|$ is an onto function

Sol:



W.E.3: $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = |x|$ is an into function

Sol:



➤ **Bijection (or) one-one & onto function:**

If $f : A \rightarrow B$ is both an injection and a surjection then f is said to be bijection or one to one and onto from A to B .

- If A, B are finite sets and $f : A \rightarrow B$ is a bijection then $n(A) = n(B)$.
- Identity function on any non empty set A is bijection.
- If A, B are finite sets and $n(A) = n(B)$ then number of bijective functions defined from A to B is $(n(A))!$
- A constant function is bijection if $n(A) = n(B) = 1$

v) **Horizontal line test:** If every straight line parallel to x-axis from the points in codomain intersects the graph of the function at only one point then f is bijection.

➤ **Equality of Functions:** The functions f and g are said to be equal if,

- the domain of f = the domain of g
- $f(x) = g(x)$ for every x in domain.

➤ **Constant function:** A function $f : A \rightarrow B$ is a constant function if the range of f contains only one element.

➤ The number of constant functions from A to B is $n(B)$.

➤ The graph of constant function is a line parallel to x-axis.

➤ Range of any constant function is a singleton set.

➤ **Identity function:** Let A be a non - empty set then $f : A \rightarrow A$ defined by $f(x) = x$, $\forall x \in A$ is called the identity function on A and it is denoted by I_A .

➤ The graph of identity function is a straight line passing through origin and inclined at an angle of 45° with x-axis.

E.g: Let $A = \{1, 2, 3, 4\}$, then the identity function on A is $I_A = \{(1,1)(2,2)(3,3)(4,4)\}$

➤ **Composite function:** If $f : A \rightarrow B$ and $g : B \rightarrow C$ are functions then $g \circ f : A \rightarrow C$ is defined by $(g \circ f)(x) = g\{f(x)\} \forall x \in A$ is called the composite function of f & g .

➤ **Inverse function:** A function $f : X \rightarrow Y$ is defined to be invertible if there exists a function $g : Y \rightarrow X$ such that $g \circ f = I_X$ and $f \circ g = I_Y$. The function g is called the inverse of f , and is denoted by f^{-1} .

➤ If f is invertible then f must be bijective i.e one-one and onto.

➤ If the inverse of a function exists then it is said to be invertible. The inverse of a function if exists then it is unique.

➤ Graph of $y = f(x)$ and $y = f^{-1}(x)$ are symmetrical about $y = x$ and intersects on the line $y = x$, and also $f(x) = f^{-1}(x) = x$ when ever graph intersects.

➤ **Working Rule to find the inverse of a function**

Let $f: X \rightarrow Y$ be a bijection function.

put $f(x) = y$. Solve the equation $y = f(x)$ to obtain x in terms of y . Interchange x and y to obtain the inverse of f .

W.E.4: $f: \mathbb{R} \rightarrow (-\infty, 1)$ given by $f(x) = 1 - 2^{-x}$.

Then $f^{-1}(x)$ is

Sol: Here $f(x)$ is both one-one and onto

Let $y = 1 - 2^{-x}$ or $2^{-x} = 1 - y$ or

$-x = \log_2(1 - y)$ or

$f^{-1}(x) = g(x) = -\log_2(1 - x)$

➤ **Properties of composite function :**

i) If $f: A \rightarrow B$, $g: B \rightarrow C$ are one-one then

$g \circ f: A \rightarrow C$ is also one-one.

ii) If $f: A \rightarrow B$, $g: B \rightarrow C$ are onto then $g \circ f: A \rightarrow C$ is also onto.

iii) If $g \circ f: A \rightarrow C$ is one-one then f is one-one.

iv) If $g \circ f: A \rightarrow C$ is onto then g is onto.

v) If $f: A \rightarrow B$ is a function then

$$f \circ I_A = I_B \circ f = f.$$

vi) If $f: A \rightarrow B$ and $g: B \rightarrow C$ are bijections

then $g \circ f: A \rightarrow C$ is a bijection &

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}.$$

vii) If $f: A \rightarrow B$ is a bijection then

$$f \circ f^{-1} = I_B, f^{-1} \circ f = I_A.$$

viii) If $f: A \rightarrow B$ and $g: B \rightarrow A$ are functions such

that $g \circ f = I_A$ and $f \circ g = I_B$ then $g = f^{-1}$.

ix) If $f: A \rightarrow A$ is a bijection then

$$f \circ f^{-1} = f^{-1} \circ f = I_A.$$

x) In general $f \circ g \neq g \circ f$.

xi) If $f: A \rightarrow B$, $g: B \rightarrow C$ and $h: C \rightarrow D$ are functions then $h \circ (g \circ f) = (h \circ g) \circ f$

➤ **Even & odd functions:**

$f: A \rightarrow \mathbb{R}, A \subseteq \mathbb{R}$ is said to be an even function if

$$f(-x) = f(x), \forall x \in A.$$

E.g: $y = |x|$, $y = x^2$, $y = \cos x$ are some even functions.

➤ **Note:** (i) The graph of an even function is symmetric about Y-axis.

(ii) If (x, y) is point on the even function then $(-x, y)$ is also point on even function.

➤ $f: A \rightarrow \mathbb{R}, A \subseteq \mathbb{R}$ is said to be an odd function if $f(-x) = -f(x) \quad \forall x \in A$.

E.g: $y = x$, $y = x^3$, $y = \sin x$ are some odd functions.

➤ **Note:** (i) The graph of an odd function is symmetric about origin (symmetrical in opposite quadrants)

(ii) If (x, y) is a point on an odd function graph then $(-x, -y)$ is also a point on the same odd function graph.

➤ **Important points of odd and even functions:**

➤ (i) A function which is even or odd, when even power is always even function.

(ii) The derivative of an odd function is an even function and derivative of an even function is an odd function.

(iii) Every function can be uniquely expressed as the sum of an even and an odd function. i.e.,

$$f(x) = \frac{1}{2}\{f(x) + f(-x)\} + \frac{1}{2}\{f(x) - f(-x)\} \\ = \{\text{even function}\} + \{\text{odd function}\}$$

➤ **Note:** (i) A function may be neither even nor odd. for example $f(x) = \sin x + x^2$ is neither even nor odd.

(ii) $f(x) = 0$ is the function which is both even and odd.

(iii) $f(x) = c$ is an even function

(iv) Every even function $y = f(x)$ is not one-one $\forall x \in D_f$.

W.E.5: Identify whether the function

$f(x) = xg(x)g(-x) + \tan(\sin x)$ is odd or even.

Sol: $f(-x) = (-x)g(-x).g(x) + \tan(\sin(-x)) \\ = -(xg(x)g(-x) + \tan(\sin x)) = -f(x)$

Hence $f(x)$ is an odd function.

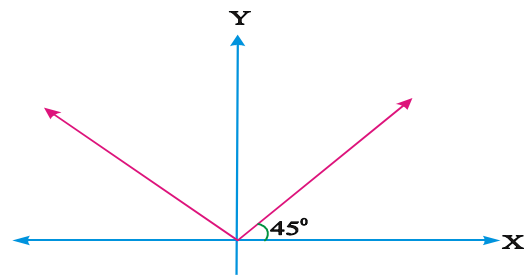
- **Polynomial function:** If $f: R \rightarrow R$ is defined by
 $f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_{n-1}x + a_n$,
 $a_0 \neq 0, a_1, a_2, \dots, a_n \in R$, n is a non-negative integer is a polynomial function of degree n in x
E.g: (i) $x^4 - x + 2$ is a polynomial function.
 (ii) $x^4 - \sqrt{x} + 2$ is not a polynomial function
- **Rational Function:** A function of the form $\frac{f(x)}{g(x)}$ where $f(x)$ and $g(x)$ are polynomial functions and $g(x) \neq 0$ is called a rational function.
- **Algebraic functions:** A function f is said to be an algebraic function if it arises due to a finite number of fundamental operations like addition, subtraction, multiplication, division and root extraction etc, on polynomial functions.
- **Transcendental function:** The functions which are not algebraic are called transcendental functions.
E.g: i) Exponential functions
 ii) Logarithmic functions
 iii) Trigonometric functions
 iv) inverse trigonometric functions
- E.g:** $f(x) = x^3 + 3x^2 - \sqrt{x^2 - x} + \sqrt[3]{x}$ is an algebraic function where as
 $g(x) = x + \sin^{-1} x$.
 $h(x) = e^x \cdot \cosh^{-1} x$ are transcendental functions.
- **Algebra of real valued functions :**
- **Definition :** Let $f: A \rightarrow B$ is a function and
 i) if $A \subseteq R$ then f is called a real variable function.
 ii) if $B \subseteq R$ then f is called a real valued function.
 iii) if $A \subseteq R, B \subseteq R$ then f is called a real function.
- **Properties:** If f and g are real valued functions with domain A and B respectively, then both f and g are defined on $A \cap B$ when $A \cap B \neq \phi$.
- Let $f: A \rightarrow R$ & $g: B \rightarrow R$ then
 (i) $(f \pm g)(x) = f(x) \pm g(x), \forall x \in A \cap B$
 (ii) $(fg)(x) = f(x) \cdot g(x), \forall x \in A \cap B$

- (iii) $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}, \forall x \in A \cap B \text{ \& } g(x) \neq 0$
 (iv) $(f \pm k)(x) = f(x) \pm k, k \in R$
 (v) $(kf)(x) = kf(x), k \in R$
 (vi) $f^n(x) = \{f(x)\}^n, n > 0$
 (vii) $|f|(x) = |f(x)|, x \in A$
- **Modulus function(Absolute value function):**
 The absolute value or numerical value or the modulus of real number x denoted by $|x|$ is defined as

$$|x| = \begin{cases} -x & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ x & \text{if } x > 0 \end{cases}$$

Thus we have $|x| \geq 0$ and $|-x| = |x|$

- The graph of $f(x) = |x|$ is



- The domain of $|x|$ is R and range is $[0, \infty)$

- **Properties of modulus function:**

- (i) $|x| \leq a \Rightarrow -a \leq x \leq a; (a \geq 0)$
 (ii) $|x| \geq a \Rightarrow x \leq -a \text{ and } x \geq a; (a \geq 0)$
 (iii) $|x + y| = |x| + |y| \Leftrightarrow x, y \geq 0 \text{ or } x, y \leq 0$
 (iv) $|x - y| = |x| - |y| \Rightarrow x \geq 0, y \geq 0, |x| \geq |y|$

(or)

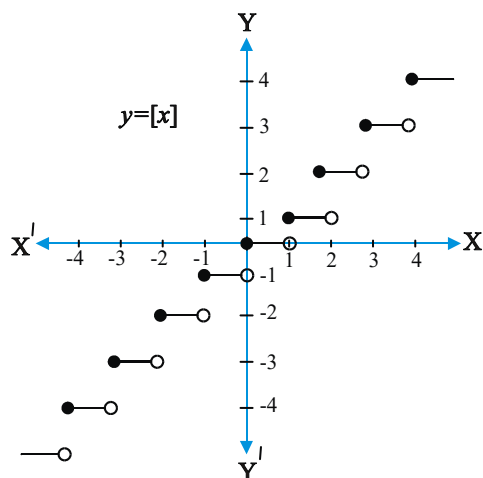
$$x \leq 0, y \leq 0, |x| \geq |y|$$

- (v) $|x \pm y| \leq |x| + |y|$ (vi) $|x \pm y| \geq ||x| - |y||$

- **Step function(Greatest integer function) (or) floor function :**

The function $f: R \rightarrow R$ defined by $f(x) = [x]$ is called the greatest integer function, where $[x]$ equal to integral part of x or greatest integer less than or equal to x . (or) If x is any real number then there exist integers n and $n+1$ such that

$n \leq x < n+1$. Then the integral part of x is defined as n . It is denoted by $[x]$.



➤ From the definition of $[x]$, we have

(i) $[x] = -1$, for $-1 \leq x < 0$

(ii) $[x] = 0$, for $0 \leq x < 1$

(iii) $[x] = 1$, for $1 \leq x < 2$

(iv) $[x] = 2$, for $2 \leq x < 3$

➤ The domain of $[x]$ is $\mathbb{R}$ and range is $\mathbb{Z}$

➤ **Properties of greatest integer function:**

(i) If $f(x) = [x + n] = [x] + n$, where $n \in \mathbb{I}$ and $[.]$ denotes the greatest integer function, then $f(x) = n + [x]$

(ii) $x - 1 < [x] \leq x$

(iii) $[x] + [-x] = \begin{cases} 0 & \text{if } x \in \mathbb{I} \\ -1 & \text{if } x \notin \mathbb{I} \end{cases}$

(iv) $[x] - [-x] = 2n - 1$, if $n - 1 < x < n$, $n \in \mathbb{I}$

(v) $[x] - [-x] = 2n + 1$, if $x = n + \{x\}$, $n \in \mathbb{I}$

(vi) $[x + y] \geq [x] + [y]$.

(vii) $[x] \geq k \Rightarrow x \geq k$, where $k \in \mathbb{Z}$

(viii) $[x] \leq k \Rightarrow x < k + 1$, where $k \in \mathbb{Z}$

(ix) $[x] > k \Rightarrow x \geq k + 1$, where $k \in \mathbb{Z}$

(x) $[x] < k \Rightarrow x < k$, where $k \in \mathbb{Z}$

(xi) $[x] \leq x < [x] + 1$

(xii) $[x + y] = [x] + [y]$ if either x (or) y (or) both are integers

(xiii) $[[x]] = [x]$

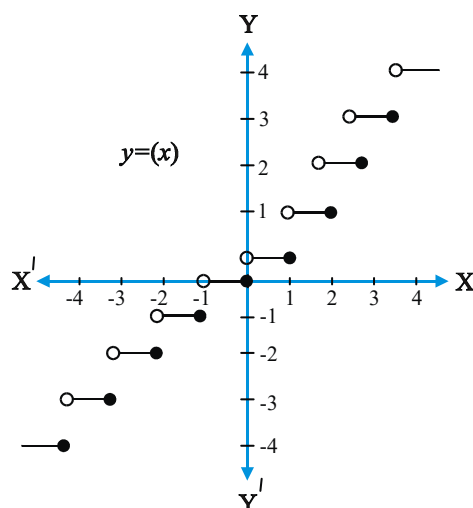
(xiv) $\left[\frac{x}{n}\right] + \left[\frac{x+1}{n}\right] + \left[\frac{x+2}{n}\right] + \dots + \left[\frac{x+n-1}{n}\right] = [x], n \in \mathbb{N}$

(xv) $[x] + \left[x + \frac{1}{n}\right] + \left[x + \frac{2}{n}\right] + \dots + \left[x + \frac{n-1}{n}\right] = [nx]$

➤ **Least Integer Function:**

If $x \in \mathbb{R}$ then the least integer greater than or equal to x is denoted by (x) is called least integer function or ceiling function.

Fig: (i) $(3.2) = 4$, (ii) $(-2.3) = -2$, (iii) $(5) = 5$



i) $(x) = \begin{cases} -1, & -2 < x \leq -1 \\ 0, & -1 < x \leq 0 \\ 1, & 0 < x \leq 1 \\ 2, & 1 < x \leq 2 \end{cases}$

ii) Domain of (x) is $\mathbb{R}$ and Range is $\mathbb{Z}$.

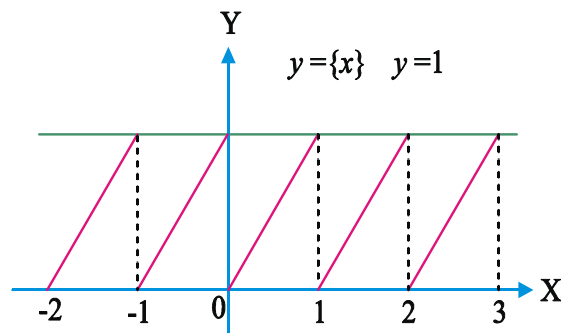
iii) If $k \in \mathbb{Z}$ and $x \in \mathbb{R}$ then $(x + k) = (x) + k$

iv) If $x \in \mathbb{Z}$, $(-x) = -x$ and

if $x \notin \mathbb{Z}$ then $(-x) = -(x) + 1$

➤ **Fractional Part of x:**

For $x \in \mathbb{R}$, the function $f(x) = x - [x]$ is called fractional part of x , it is denoted by $\{x\}$ or $\{x\}$.



- Domain of $\{x\}$ is $\mathbb{R}$ and Range of $\{x\}$ is $[0,1)$
- The function $\{x\}$ is neither even nor odd

$$\{x\} = \begin{cases} x+2, & -2 \leq x < -1 \\ x+1, & -1 \leq x < 0 \\ x, & 0 \leq x < 1 \\ x-1, & 1 \leq x < 2 \\ x-2, & 2 \leq x < 3 \end{cases}$$

i) If x is an integer then $x = [x]$ and $\{x\} = 0$

ii) $\{[x]\} = 0$

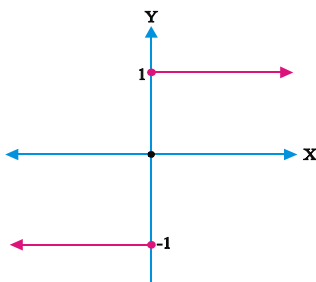
$$\text{iii) } \{x\} + \{-x\} = \begin{cases} 0, & \text{if } x \in \mathbb{Z} \\ 1, & \text{if } x \notin \mathbb{Z} \end{cases}$$

iv) $x = [x] + \{x\}$, $[.]$ denote the integral part and $\{.\}$ fractional part of x respectively

- **Signum Function:** The signum function or signature function is defined as

$$\text{Sgn}(x) = \begin{cases} \frac{|x|}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

$$\text{i.e. } \text{Sgn}(x) = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \\ 0 & \text{if } x = 0 \end{cases}$$



- The domain of $\text{Sgn}(x)$ is $\mathbb{R}$ and its Range is $\{-1, 0, 1\}$
- **Linear function:** The function f defined by $f(x) = mx + c$, $x \in \mathbb{R}$, is called linear function, where m and c are constants.
- **Explicit function:** A function $y = f(x)$ is said to be an explicit function of x if the dependent

variable y can be expressed in terms of the independent variable x .

E.g: i) $y = x - \cos x$, ii) $y = x + \log_e x - 2x^3$.

- **Implicit function:** A function $y = f(x)$ is said to be an implicit function of x if the dependent variable y can not be expressed in terms of the independent variable x only.

E.g: i) $3x^2 - 4x^2y^2 + 2xy = 7$ ii) $xy = \cos(x+y)$

- **Periodic function:** A function $f(x)$ is said to be periodic function if there exists a positive real number T , such that $f(x+T) = f(x)$, $\forall x \in D_f$, where the least positive real number T is called its fundamental period.

W.E.6: The period (if periodic) of the function

$f(x) = x - [x - b]$, $b \in \mathbb{R}$ ($[.]$ denotes the greatest integer function) is

Sol: $f(x) = x - [x - b]$

$$x - b - [x - b] + b = b + \{x - b\}$$

So, $f(x)$ has period 1. ($\because$ Period $\{x\}$ is 1)

- **Extension of domains:** Let a function be defined on certain domain which is entirely non-negative (or non-positive). The domain of

$f(x)$ can be extended to the set

$X = \{-x; x \in \text{domain of } f(x)\}$ in two ways :

Even extension: The even extension is obtained by defining a new function $f(-x)$ for $x \in X$, such that $f(-x) = f(x)$.

Odd extension: The odd extension is obtained by defining a new function $f(-x)$ for, such that $f(-x) = -f(x)$.

W.E.7: If $f(x) = \begin{cases} x^3 + x^2 & \text{for } 0 \leq x \leq 2 \\ x + 2 & \text{for } 2 < x \leq 4 \end{cases}$

Then the even and odd extension of $f(x)$ is

Sol: The even extension of $f(x)$ is as follows :

$$g(x) = \begin{cases} -x + 2, & -4 \leq x < -2 \\ -x^3 + x^2, & -2 \leq x \leq 0 \end{cases}$$

The odd extension of $f(x)$ is as follows :

$$h(x) = \begin{cases} x-2, & -4 \leq x < -2 \\ x^3 - x^2, & -2 \leq x \leq 0 \end{cases}$$

W.E.8: The domain of $f(x) = \sqrt{\sin^{-1}(\log_2 x)}$ is

Sol: $\sin^{-1}(\log_2 x) \geq 0$ and $-1 \leq \log_2 x \leq 1$ and $x > 0$
 $\log_2 x \geq 0$ & $\log_2 x \leq 1$ & $x > 0$.
 $0 \leq \log_2 x \leq 1, x > 0$
 $2^0 \leq x \leq 2^1 \quad \therefore x \in [1, 2]$

W.E.9: The range of $f(x) = \sin^{-1} x + \tan^{-1} x$ is

Sol: Domain of $f(x)$ is $[-1, 1]$

$$\sin^{-1} x \in \left[-\frac{\pi}{2}, \frac{\pi}{2} \right], \tan^{-1} x \in \left[-\frac{\pi}{4}, \frac{\pi}{4} \right]$$

$$-\frac{\pi}{2} - \frac{\pi}{4} \leq \sin^{-1} x + \tan^{-1} x \leq \frac{\pi}{2} + \frac{\pi}{4}$$

$$-\frac{3\pi}{4} \leq f(x) \leq \frac{3\pi}{4}$$

➤ **To find the range:**

i) If the domain of $f(x)$ has finite number of elements then Range is equal to set of corresponding values of $f(x)$.

W.E.10: The range of $f: A \rightarrow B$ where

$A = \{1, 2, 3, 4\}$ and $f(x)$ is defined as

$$f(x) = x^2 + x + 1 \text{ is}$$

Sol: $f(1) = 1 + 1 + 1 = 3$

$$\text{Similarly } f(2) = 7, f(3) = 13, f(4) = 21$$

$$\text{Range} = \{3, 7, 13, 21\}$$

ii) If the domain of $y = f(x)$ is $\mathbb{R}$ (i.e. the set of real numbers) or $\mathbb{R} - \{\text{some finite points}\}$ or an infinite interval, express x in terms of y . From this,

(a) find y for x to be defined or real, or
 (b) If a quadratic equation is formed in terms of x then apply the condition of real roots. ($\Delta \geq 0$)

W.E.11: The Range of $f(x) = \frac{x^2 + x - 3}{x - 1}$ is

Sol: $f(x) = \frac{x^2 + x - 3}{x - 1} = y \Rightarrow xy - y = x^2 + x - 3$

$$x^2 + x(1 - y) + y - 3 = 0$$

Quadratic eq. in terms of x and $x \in \mathbb{R}$

$$\therefore b^2 - 4ac \geq 0, (1 - y)^2 - 4(y - 3)(1) \geq 0$$

$$(y - 3)^2 + 4 \geq 0 \quad y \in \mathbb{R} \quad \therefore \text{range is } \mathbb{R}.$$

W.E.12: The range of the following functions defined over $[0, 3]$

$$f(x) = \begin{cases} x; & 0 \leq x \leq 1 \\ x^2 + 1; & 1 < x \leq 2 \\ x^3; & 2 < x \leq 3 \end{cases} \text{ is}$$

Sol: Obviously the domain of the function is $[0, 3]$.

Its range is $[0, 1]$ for $0 \leq x \leq 1$;

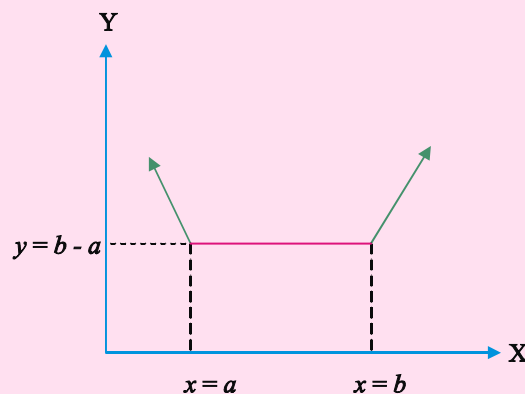
$(2, 5]$ for $1 < x \leq 2$ and $(8, 27]$ for $2 < x \leq 3$.

$$\therefore \text{Range} = [0, 1] \cup (2, 5] \cup (8, 27].$$

iii) If the graph of the function is known, then the range of the function can be obtained easily from the graph.

W.E.13: The range of $|x - a| + |x - b|$ ($b > a$) is

Sol:



from figure we conclude range of $f(x)$ is $[b - a, \infty)$.

iv) By using $A.M \geq G.M \geq H.M$ concept.

W.E.14: The range of the function

$$f(x) = 6^x + 3^x + 6^{-x} + 3^{-x} + 2 \text{ is}$$

Sol: $6^x + 6^{-x} \geq 2\sqrt{6^x \cdot 6^{-x}} = 2$

$$3^x + 3^{-x} \geq 2\sqrt{3^x \cdot 3^{-x}} = 2$$

$$\therefore f(x) = 6^x + 3^x + 6^{-x} + 3^{-x} + 2 \geq 2 + 2 + 2,$$

$$f(x) \geq 6$$

$\therefore$ range of $f(x)$ is $[6, \infty)$

v) Range of $f(x) = ax^2 + bx + c$ is $\left[-\frac{D}{4a}, \infty\right)$

if $a > 0$ and $\left(-\infty, -\frac{D}{4a}\right]$ if $a < 0$,

where $D = b^2 - 4ac$.

vi) If $a^2 + b^2 + c^2 = k$, then range of

$ab + bc + ca$ is $\left[-\frac{k}{2}, k\right]$

➤ **Domain and range of some standard functions:**

S.No	Function	Domain	Range
1.	$\sqrt{a^2 - x^2}$	$[-a, a]$	$[0, a]$
2.	$\frac{1}{\sqrt{a^2 - x^2}}$	$(-a, a)$	$\left[\frac{1}{a}, \infty\right)$
3.	$\sqrt{x^2 - a^2}$	$R - (-a, a)$	$[0, \infty)$
4.	$\frac{1}{\sqrt{x^2 - a^2}}$	$R - [-a, a]$	$(0, \infty)$
5.	$a^x, (a > 0, a \neq 1)$	R	$(0, \infty)$
6.	e^x	R	$(0, \infty)$
7.	$\log_a x$ ($a > 0, a \neq 1$)	$(0, \infty)$	R
8.	$ x $	R	$[0, \infty)$
9.	$[x]$	R	Z
10.	$\{x\} = x - [x]$	R	$[0, 1)$
11.	$\sqrt{x}$	$[0, \infty)$	$[0, \infty)$
12.	$\sin x$	R	$[-1, 1]$
13.	$\cos x$	R	$[-1, 1]$
14.	$\tan x$	$R - \left\{(2n+1)\frac{\pi}{2} : n \in Z\right\}$	R
15.	$\operatorname{cosec} x$	$R - \{n\pi : n \in Z\}$	$R - (-1, 1)$
16.	$\sec x$	$R - \left\{(2n+1)\frac{\pi}{2} : n \in Z\right\}$	$R - (-1, 1)$
17.	$\cot x$	$R - \{n\pi : n \in Z\}$	R
18.	$\sin^{-1} x$	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
19.	$\cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
20.	$\tan^{-1} x$	R	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

21.	$\operatorname{Cosec}^{-1} x$	$R - (-1, 1)$	$\left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$
22.	$\sec^{-1} x$	$R - (-1, 1)$	$\left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$
23.	$\cot^{-1} x$	R	$(0, \pi)$
24.	$a \cos x + b \sin x + c$	R	$\left[c - \sqrt{a^2 + b^2}, c + \sqrt{a^2 + b^2}\right]$
25.	$\sinh x$	R	R
26.	$\cosh x$	R	$[1, \infty)$
27.	$\tanh x$	R	$(-1, 1)$
28.	$\operatorname{cosech} x$	$R - \{0\}$	$R - \{0\}$
29.	$\operatorname{sech} x$	R	$(0, 1]$
30.	$\coth x$	$R - \{0\}$	$R - [-1, 1]$
31.	$\sinh^{-1} x$	R	R
32.	$\cosh^{-1} x$	$[1, \infty)$	$[0, \infty)$
33.	$\tanh^{-1} x$	$(-1, 1)$	R
34.	$\operatorname{cosech}^{-1} x$	$R - \{0\}$	$R - \{0\}$
35.	$\operatorname{sech}^{-1} x$	$(0, 1]$	$[0, \infty)$
36.	$\coth^{-1} x$	$R - [-1, 1]$	$R - \{0\}$

➤ **Functional equations:** If $f(x)$ is a function such that

i) $f(x+y) = f(x)f(y)$ then $f(x) = k^x$

($k \in R^+$)

ii) $f(x+y) = f(x) + f(y)$ then $f(x) = kx$

iii) $f(xy) = f(x) + f(y)$ then

$f(x) = k \log_a x$ ($a \neq 1, a > 0$)

iv) If $f(x)$ is a polynomial function such that

$f(x) + f\left(\frac{1}{x}\right) = f(x)f\left(\frac{1}{x}\right)$ then $f(x) = 1 \pm x^n$.

v) If $f(x)$ is a function such that

$f(x+y) + f(x-y) = 2f(x)f(y)$

then $f(x) = \frac{k^x + k^{-x}}{2}$ or $f(x) = \cos x$



1. $f : A \rightarrow B$ is a function then **A, B are respectively.**

- 1) domain, range 2) domain, co domain
3) co domain, range 4) range, domain

2. $f : A \rightarrow B$ then **$f(A)$ is called**

- 1) domain 2) co domain 3) range 4) function

3. If $f: A \rightarrow B$ is a function then
 1) $f(A) = B$ 2) $f(A) \subset B$
 3) $f(A) \subseteq B$ 4) $B \subseteq f(A)$
4. If $f: A \rightarrow B$ is surjective then
 1) No two elements of A have the same image in B
 2) Every element in A has an image in B
 3) Every element of B has at least one pre-image in A
 4) A and B are finite non empty sets
5. A constant function $f: A \rightarrow B$ will be one-one if
 1) $n(A) = n(B)$ 2) $n(A) = 1$
 3) $n(B) = 1$ 4) $n(A) < n(B)$
6. If $f: A \rightarrow B$ is a constant function which is onto then B is
 1) a singleton set 2) a null set
 3) an infinite set 4) a finite set
7. If $n \geq 2$ then the number of surjections that can be defined from $\{1, 2, 3, \dots, n\}$ onto $\{1, 2\}$ is
 1) $n^2 - n$ 2) n^2 3) 2^n 4) $2^n - 2$
8. If f and g are functions such that $f \circ g$ is onto then
 1) f is onto 2) g is onto
 3) $g \circ f$ is onto 4) Neither f nor g is onto
9. If f and g are functions such that $f \circ g$ is one - one then
 1) g must be onto 2) g must be one - one
 3) f must be one-one 4) f, g need not one-one
10. To have inverse for the function f, f should be
 1) one-one 2) onto
 3) one-one and onto 4) Identity function
11. If $f: A \rightarrow B$ is a bijection then $f^{-1} \circ f =$
 1) $f \circ f^{-1}$ 2) f 3) f^{-1} 4) I_A
12. $f: A \rightarrow B$ is a bijection then $f \circ f^{-1} =$
 1) I_A 2) I_B 3) f 4) f^{-1}
13. Let $f(x) = ax^2 + bx + c$, where a, b, c are rational and $f: Z \rightarrow Z$ where Z is the set of integers. Then $a + b$ is
 1) a negative integer 2) an integer
 3) non integral rational number
 4) Real number
14. The graph of the function $y = f(x)$ is symmetrical about the line $x = 2$. Then
 1) $f(x+2) = f(x-2)$ 2) $f(2+x) = f(2-x)$
 3) $f(x) = f(-x)$ 4) $f(x) = -f(-x)$
15. The graph of odd function is
 1) symmetric about origin
 2) symmetric about y-axis
 3) parallel to x-axis 4) parallel to y-axis
16. The graph of even function is
 1) symmetric in opp quadrants
 2) symmetric about y-axis
 3) parallel to x-axis 4) parallel to y-axis
17. The graph of Identify function is
 1) Straight line passing through origin
 2) symmetric about y-axis
 3) parallel to x-axis 4) parallel to y-axis
18. The graph of $|x|$ is
 1) Straight line passing through origin
 2) symmetric about y-axis
 3) parallel to x-axis 4) parallel to y-axis
19. The function $f(x) = \{x\}$ is
 1) even function 2) odd function
 3) periodic function with period 1
 4) periodic function but no fundamental period
20. Let f be a strictly decreasing function with range $[a, b]$ then domain of the function f^{-1} is
 1) $[f^{-1}(b), f^{-1}(a)]$ 2) $[b, a]$
 3) $[f^{-1}(a), f^{-1}(b)]$ 4) (b, a)
21. If $f(x) = x^2 + \lambda x + \mu$ be an integral function of the integral variable x then
 1) λ is an integer and μ is a rational fraction
 2) λ and μ are integers
 3) μ is an integer and λ is a rational fraction
 4) λ and μ are rational fractions

C.U.Q-KEY

- 01) 2 02) 3 03) 3 04) 3 05) 2 06) 1
 07) 4 08) 1 09) 2 10) 3 11) 4 12) 2
 13) 2 14) 2 15) 1 16) 2 17) 1 18) 2
 19) 3 20) 1 21) 2



LEVEL - I (C.W)

REAL VALUED FUNCTIONS

1. Let $f: \{(1,1), (2,3), (0,-1), (-1,-3)\}$ be a function from Z to Z defined by $f(x) = ax + b$, for some integers a, b then $(a, b) =$
 1) $(-1, 2)$ 2) $(2, -1)$ 3) $(3, -2)$ 4) $(0, 3)$

2. If $e^{f(x)} = \frac{10+x}{10-x}, x \in (-10, 10)$
and $f(x) = k \cdot f\left(\frac{200x}{100+x^2}\right)$ then $k =$
1) 0.5 2) 0.6 3) 0.7 4) 0.8
3. $f: R \rightarrow R$ is defined as $f(x) = 2x + |x|$ then
 $f(3x) - f(-x) - 4x =$ (EAM-97)
1) $f(x)$ 2) $-f(x)$ 3) $f(-x)$ 4) $2f(x)$
4. $f(1) = 1, n \geq 1 \Rightarrow f(n+1) = 2f(n) + 1$
then $f(n) =$ (EAM-92)
1) 2^{n+1} 2) 2^n 3) $2^n - 1$ 4) $2^{n-1} - 1$
5. If $f(x) = \frac{7^{1+\ln x}}{x^{\ln 7}}$ then $f(2015) =$
1) 20 2) 7 3) 2015 4) 100
6. If $f(x) = \frac{\cos^2 x + \sin^4 x}{\sin^2 x + \cos^4 x}$ for $x \in R$ then
 $f(2016) =$
1) 1 2) 2 3) 3 4) 4
7. If $f = \{(-2, 4), (0, 6), (2, 8)\}$ and
 $g = \{(-2, -1), (0, 3), (2, 5)\}$, then
 $\left(\frac{2f}{3g} + \frac{3g}{2f}\right)(0) =$
(1) $1/12$ (2) $25/12$ (3) $5/12$ (4) $13/12$
8. If $f(x) = \sin(\log x)$ then
 $f(xy) + f\left(\frac{x}{y}\right) - 2f(x) \cos(\log y) =$
1) $\cos(\log x)$ 2) $\sin(\log y)$
3) $\cos(\log(xy))$ 4) 0
9. If $f(x+y, x-y) = xy$ then the arithmetic
mean of $f(x, y)$ and $f(y, x)$ is
1) x 2) y 3) 0 4) xy

EVEN AND ODD FUNCTIONS

10. Let $f(x) = \frac{x}{e^x - 1} + \frac{x}{2} + 1$, then f is
1) an odd function 2) an even function
3) both odd and even 4) neither odd nor even
11. Which of the following is an even function
1) $f(x) = \frac{a^x + a^{-x}}{a^x - a^{-x}}$ 2) $f(x) = \frac{a^x + 1}{a^x - 1}$
3) $f(x) = x \frac{a^x - 1}{a^x + 1}$ 4) $f(x) = \log_2(x + \sqrt{x^2 + 1})$

PERIODIC FUNCTIONS

12. The period of $\cos x^2$ is
1) 2π 2) $\sqrt{2}\pi$ 3) $4\pi^2$ 4) does not exist
13. Period of $f(x) = e^{\cos\{x\}} + \sin \pi[x]$ is $([\cdot])$
and $\{\cdot\}$ denote the greatest integer
function and fractional part function
respectively)
1) 1 2) 2 3) π 4) 2π
14. Let $f(x)$ be periodic and k be a positive
real number such that $f(x+k) + f(x) = 0$
for all $x \in R$. Then the period of $f(x)$ is
1) k 2) $2k$ 3) $4k$ 4) $8k$
15. The period of $f(x) = \sqrt{x - [x]}$ is
1) no fundamental period 2) $\frac{1}{2}$
3) 1 4) 2

DOMAIN OF THE FUNCTION

16. The domain of $f(x) = \frac{x^2 + 2x + 1}{x^2 - x - 6}$
1) $R - \{3, -2\}$ 2) $R - \{-3, 2\}$
3) $R - \{3, -2\}$ 4) $R - (3, -2)$
17. The domain of $f(x) = \frac{1}{\log|x|}$ is
1) $R - \{0\}$ 2) $R - \{0, 1\}$
3) $R - \{-1, 0, 1\}$ 4) $(-\infty, \infty)$
18. The domain of $f(x) = \frac{3^x + 3^{-x}}{3^x - 3^{-x}}$ is
1) $(-\infty, \infty)$ 2) $(-\infty, 0) \cup (0, \infty)$
3) $(0, \infty)$ 4) $(0, 1)$
19. The domain of the function $f(x) = \sqrt{\log_{16} x^2}$ is
1) $x = 0$ 2) $|x| \geq 4$ 3) $|x| \geq 1$ 4) $|x| \geq 2$
20. The domain of $f(x) = \frac{1}{\sqrt{|x| - x}}$ is [AIE-2011]
1) $(-\infty, 0)$ 2) $(0, \infty)$ 3) $(1, \infty)$ 4) $(-\infty, \infty)$

21. The domain of $f(x) = \frac{1}{[x]-x}$ is
1) R 2) Z 3) R - Z 4) Q - {0}
22. The domain of $f(x) = \sqrt{x-2} + \frac{1}{\log(4-x)}$ is
1) $[2, \infty)$ 2) $(-\infty, 4)$
3) $[2, 3) \cup (3, 4)$ 4) $[3, \infty)$
23. The domain of $f(x) = e^{\sqrt{x}} + \cos x$ is
1) $(-\infty, \infty)$ 2) $[0, \infty)$ 3) $(0, 1)$ 4) $(1, \infty)$
24. The domain of $\log_a \sin^{-1} x$ is ($a > 0, a \neq 1$)
1) $0 < x \leq 1$ 2) $0 \leq x \leq 1$
3) $0 \leq x < 1$ 4) $0 < x < 1$
25. The domain of $\cosh^{-1} 5x$ is
1) R 2) $[0, \infty)$ 3) $\left(\frac{1}{5}, \infty\right)$ 4) $\left[\frac{1}{5}, \infty\right)$
26. For which Domain, the functions
 $f(x) = 2x^2 - 1$ and $g(x) = 1 - 3x$ are equal to
1) R 2) $\left\{\frac{1}{2}, -2\right\}$ 3) $\left(\frac{1}{2}, 2\right)$ 4) $\left[\frac{1}{2}, 2\right]$
- RANGE OF THE FUNCTION**
27. The domain and range of the real function f defined by $f(x) = \frac{4-x}{x-4}$ is given by
1) Domain = R, Range = $\{-1, 1\}$
2) Domain = $R - \{1\}$, Range = R
3) Domain = $R - \{4\}$, Range = $\{-1\}$
4) Domain = $R - \{-4\}$, Range = $\{-1, 1\}$
28. Range of $f(x) = \frac{1}{1-2\cos x}$ is
1) $\left[\frac{1}{3}, 1\right]$ 2) $\left[-1, \frac{1}{3}\right]$
3) $(-\infty, -1] \cup \left[\frac{1}{3}, \infty\right)$ 4) $\left[-\frac{1}{3}, 1\right]$
29. The range of $f(x) = x^2 + x + 1$ is
1) $\left[\frac{3}{4}, \infty\right)$ 2) $[0, \infty)$ 3) $[1, \infty)$ 4) $\left[\frac{1}{4}, \infty\right)$
30. The domain and range of the function f given by $f(x) = 2 - |x - 5|$ is
1) Domain = R^+ , Range = $(-\infty, 1]$
2) Domain = R, Range = $(-\infty, 2]$
3) Domain = R, Range = $(-\infty, 2)$
4) Domain = R^+ , Range = $(-\infty, 2]$

31. $f = \{(x, \frac{x^2}{x^2+1}) : x \in R\}$, be a function R into R, range of 'f'
1) $[0, 1)$ 2) $(-\infty, \infty)$ 3) $(0, \infty)$ 4) R^+
32. Range of the function $f(x) = \sqrt{[x]-x}$ is
1) R 2) $\{1\}$ 3) $\{0\}$ 4) $(0, \infty)$
33. Let $A = \{9, 10, 11, 12, 13\}$ and $f : A \rightarrow N$ be defined by $f(n) =$ highest prime factor of n , then its range is
1) $\{13\}$ 2) $\{3, 5, 11, 13\}$
3) $\{11, 13\}$ 4) $\{2, 3, 5, 11\}$
34. The range of $f(x) = \frac{x^2 - x + 1}{x^2 + x + 1}$ is
1) $\left[\frac{1}{3}, 3\right]$ 2) $\left[\frac{1}{2}, 2\right]$ 3) $[0, 1]$ 4) $[-1, 1]$

TYPES OF FUNCTIONS

35. The equivalent function of $\log x^2$ is
1) $2 \log x$ 2) $2 \log |x|$
3) $|\log x^2|$ 4) $(\log x)^2$
36. The number of linear functions which map $[-1, 1]$ to $[0, 2]$ are
1) One 2) Two 3) Four 4) Three
37. If $A = (3, 81)$ and $f : A \rightarrow B$ is a surjection defined by $f(x) = \log_3 x$ then $B =$
1) $[1, 4]$ 2) $(1, 4]$ 3) $(1, 4)$ 4) $[1, \infty)$
38. Let $f(x) = \sin^2 \frac{x}{2} + \cos^2 \frac{x}{2}$ and $g(x) = \sec^2 x - \tan^2 x$. The two functions are equal over the set
1) ϕ 2) R
3) $R - \left\{x \mid x = (2n+1)\frac{\pi}{2}, n \in Z\right\}$ 4) $R - \{0\}$
39. $f : R \rightarrow R$ defined by $f(x) = \frac{x}{x^2+1}, \forall x \in R$ is
1) one - one 2) onto
3) bijective 4) neither one one nor onto

40. If $f : Z \rightarrow Z$ is such that $f(x) = 6x - 11$ then f is
 1) injective but not surjective
 2) surjective but not injective 3) bijective
 4) neither injective nor surjective
41. $f : N \rightarrow A$ Where $A = \{0, 1\}$ defined by

$$f(x) = \begin{cases} 0 & \text{if } x \text{ is odd} \\ 1 & \text{if } x \text{ is even} \end{cases}$$
 Then f is
 (1) one - one, onto (2) one-one, into
 (3) many-one, onto (4) many-one, into
42. $f : (-\infty, \infty) \rightarrow (0, 1]$ defined by

$$f(x) = \frac{1}{x^2 + 1}$$
 is
 1) one-one but not onto
 2) onto but not one-one
 3) bijective 4) neither one-one nor onto
43. The function $f : R \rightarrow R$ defined by

$$f(x) = 4^x + 4^{|x|}$$
 is
 1) One - one and into 2) Many - one and into
 3) One - one and onto 4) Many-one and onto
- Number of functions :**
44. The number of one-one functions that can be defined from $A = \{4, 8, 12, 16\}$ to B is 5040, then $n(B) =$
 1) 7 2) 8 3) 9 4) 10
45. If $A = \{1, 8, 11, 14, 25\}$ then the condition to define a surjection from A to B is
 1) $n(A) + n(B) = 20$ 2) $n(A) < n(B)$
 3) $n(B) \leq 5$ 4) $n(B) = 10$
46. If $A = \{1, 2, 3\}$, $B = \{1, 2\}$ then the number of functions from A to B are
 1) 6 2) 8 3) 9 4) 32
47. The number of non-bijective mappings that can be defined from $A = \{1, 2, 7\}$ to itself is
 1) 21 2) 27 3) 6 4) 9
48. Let $A = \{1, 2, 3\}$ and $B = \{a, b, c\}$. If l is number of functions from A to B and m is number of one-one functions from A to B , then
 1) l is 9 2) m is 9 3) l is 27 4) m is 16
49. The number of constant functions possible from R to B where $B = \{2, 4, 6, 8, \dots, 24\}$ are
 1) 24 2) 12 3) 8 4) 6

COMPOSITE FUNCTIONS

50. The functions $f : R \rightarrow R$, $g : R \rightarrow R$ are defined as

$$f(x) = \begin{cases} 0 & \text{when } x \text{ is rational} \\ 1 & \text{when } x \text{ is irrational} \end{cases}$$

$$g(x) = \begin{cases} -1 & \text{when } x \text{ is rational} \\ 0 & \text{when } x \text{ is irrational} \end{cases}$$

then $(f \circ g)(\pi) + (g \circ f)(e) =$ (EAM-01)

- 1) -1 2) 0 3) 1 4) 2

51. If $f(x) = (a - x^n)^{\frac{1}{n}}$ then $f \circ f(x)$ is

- 1) x 2) $a - x$ 3) x^n 4) $x^{-\frac{1}{n}}$

52. If $f(x) = \frac{x}{\sqrt{1+x^2}}$ then $f \circ f \circ f(x) =$

- 1) $\frac{x}{\sqrt{1+3x^2}}$ 2) $\frac{x}{\sqrt{1-x^2}}$
 3) $\frac{2x}{\sqrt{1+2x^2}}$ 4) $\frac{x}{\sqrt{1+x^2}}$

Inverse of a function:

53. If $f(x) = \frac{e^x + e^{-x}}{2}$ then the inverse of $f(x)$ is

- 1) $\log_e(x + \sqrt{x^2 + 1})$ 2) $\log_e \sqrt{x^2 + 1}$
 3) $\log_e(x + \sqrt{x^2 - 1})$ 4) $\log_e(x - \sqrt{x^2 - 1})$

54. If $f : \{1, 2, 3, \dots\} \rightarrow \{0, \pm 1, \pm 2, \dots\}$ is defined

$$\text{by } f(n) = \begin{cases} n/2 & \text{if } n \text{ is even} \\ -\left(\frac{n-1}{2}\right) & \text{if } n \text{ is odd} \end{cases}$$

then $f^{-1}(-100)$ is

- 1) 100 2) 199 3) 201 4) 200

55. If $f(x) = \sin^{-1}\{3 - (x-6)^4\}^{1/3}$ then $f^{-1}(x) =$

- 1) $6 + \sqrt[4]{3 + \sin^3 x}$ 2) $6 + \sqrt[4]{3 - \sin^3 x}$
 3) $6 + \sqrt[4]{3 + \sin x}$ 4) $6 + \sqrt[4]{3 - \sin x}$

LEVEL-I(C.W.)-KEY

- 01) 2 02) 1 03) 4 04) 3 05) 2 06) 1
 07) 2 08) 4 09) 3 10) 2 11) 3 12) 4
 13) 1 14) 2 15) 3 16) 1 17) 3 18) 2
 19) 3 20) 1 21) 3 22) 3 23) 2 24) 1
 25) 4 26) 2 27) 3 28) 3 29) 1 30) 2
 31) 1 32) 3 33) 2 34) 1 35) 2 36) 2
 37) 3 38) 3 39) 4 40) 1 41) 3 42) 2
 43) 1 44) 4 45) 3 46) 2 47) 1 48) 3
 49) 2 50) 1 51) 1 52) 1 53) 3 54) 3
 55) 2

LEVEL-I(C.W.)-HINTS

- solve the equations
- $f(x) = 2k f(x)$
- expand and it is $2 f(x)$
- Put $n = 1$
- $7^{\ln x} = x^{\ln 7}$
- $$f(x) = \frac{\cos^2 x + \sin^2 x (1 - \cos^2 x)}{\sin^2 x + \cos^2 x (1 - \sin^2 x)} = 1$$
- $$\left(\frac{2f}{3g} + \frac{3g}{2f} \right)(0) = \frac{2f(0)}{3g(0)} + \frac{3g(0)}{2f(0)} = \frac{25}{12}$$
- $\sin(\log x + \log y) + \sin(\log x - \log y) - 2\sin(\log x)\cos(\log y) = 0$
- Let $x + y = p, x - y = q$. Then

$$f(p, q) = \frac{p+q}{2} \cdot \frac{p-q}{2} = \frac{p^2 - q^2}{4}$$

$$\therefore f(x, y) = \frac{x^2 - y^2}{4} \text{ and } f(y, x) = \frac{y^2 - x^2}{4}$$
- $f(-x) = f(x)$
- verify that the options are even and odd
- no value of T exist such that $f(x+T) = f(x)$
- $f(x)e^{\cos\{x\}}$ period of $\{x\}$ i.e. of $x - [x]$ is 1
 $\therefore$ period of $f(x)$ is 1.
- We have, $f(x+k) + f(x) = 0 \dots (1)$ for all $x \in R$. Putting $x+k$ for x , we get

$$f(x+2k) + f(x+k) = 0 \dots (2)$$

$$[(2) - (1)] \Rightarrow f(x+2k) - f(x) = 0 \Rightarrow$$

$$f(x+2k) = f(x)$$

 This shows that $f(x)$ is periodic with period $2k$.

- $f(x) = \sqrt{\{x\}}$ period is one
- denominator not equal to zero
- $\log|x| \neq 0$ and $x \neq 0 \Rightarrow x \neq \{0, 1, -1\}$
- $3^x \neq 3^{-x}$
- $x \neq 0$ and $x^2 \geq 1 \Rightarrow |x| \geq 1$
- $|x| - x > 0$
- $[x] - x \neq 0$
- $x - 2 \geq 0, 4 - x > 0$ and $x \neq 3$
- $x \geq 0$
- $\sin^{-1}x > 0 \Rightarrow 0 < x \leq 1$
- Domain of $\cosh^{-1}x = [1, \infty)$
- $2x^2 - 1 = 1 - 3x \Rightarrow (2x - 1)(x + 2) = 0$
- denominator not equal to zero.
- $-2 \leq -2\cos x \leq 2, -1 \leq 1 - 2\cos x \leq 3$
- $\left(x + \frac{1}{2}\right)^2 + \frac{3}{4} \geq \frac{3}{4}$
- $|x|$ defines for any value
- $x^2 < x^2 + 1 \therefore \frac{x^2}{x^2 + 1} \in [0, 1)$
- Domain of $x = z, \therefore [x] - x = 0 \forall x \in Z$
- $f(9) = 3, f(10) = 5, f(11) = 11, f(12) = 3, f(13) = 13$
- $$f(x) = \frac{x^2 - x + 1}{x^2 + x + 1} = y$$

$$\Rightarrow (y-1)x^2 + (y+1)x + (y-1) = 0$$
 and x is real
- As $\log x$ is defined for only positive values of x .
 But $\log x^2$ defined for all non zero real values of x ,
 also $\log|x|$ is defined $\forall$ non zero real x . Hence
 $\log x^2$ and $2\log|x|$ are identical functions.
- Width of both interval is same, which can be mapped by these functions $y = 1 - x$ and $y = 1 + x$.
- $3 < x < 81 \Rightarrow \log_3 3 < \log_3 x < \log_3 8$
- Common domain
- $f(2) = f\left(\frac{1}{2}\right), f(1)$ has no pre image.
- linear function is one-one no pre image for $y = 0$
- $f(1) = f(3) = 0 \Rightarrow f$ is not one-one
 Range of $f = \{0, 1\} \Rightarrow f$ is onto

42. $f(-1)=f(1) \Rightarrow f$ is not one one
range = codomain $\Rightarrow f$ is onto
43. Since for different x , 4^x and $4^{|x|}$ are different positive numbers
 $\therefore f$ is one one also, f is not onto as its range $(0, \infty)$ is a proper subset of its co domain.
44. No. of one-one functions from A to B = ${}_{n(B)}P_4$
45. $n(A) \geq n(B) \Rightarrow n(B) \leq 5$
46. $n(B)^{n(A)}$
47. $3^3 - 3!$
48. The no. of functions from A to B = $3^3 = 27$
49. The no. of constant functions from A to B is $n(B)$.
50. $(f \circ g)(\pi) + g \circ f(e) = f[g(\pi)] + g[f(e)]$
 $= f(0) + g(1) = 0 - 1 = -1$
51. $f \circ f(x) = f[f(x)] = f\left[\left(a - x^n\right)^{\frac{1}{n}}\right] = x$
52. $f(f(\dots\dots n \text{ times})) = \frac{x}{\sqrt{1+nx^2}}$
53. let $f(x) = y$
54. Take $f^{-1}(x) = y$
 $\Rightarrow x = f(y) = 5y - 6 \Rightarrow y = f^{-1}(x) = \frac{x+6}{5}$
Let $f^{-1}(-100) = x \Rightarrow f(x) = -100$
 $\Rightarrow \frac{-(x-1)}{2} = -100 \Rightarrow x = 201$
55. $f^{-1}(x) = y \Rightarrow x = f(y) = \sin^{-1}(3 - (y-6)^4)^{1/3}$
 $\Rightarrow f^{-1}(x) = 6 + \sqrt[4]{3 - \sin^3 x}$.



LEVEL - I (H.W)

REAL VALUED FUNCTIONS

1. f is defined by $f(x) = \begin{cases} x^2, 0 \leq x \leq 2 \\ 3x, 2 \leq x \leq 10 \end{cases}$ then $f(2) =$
1) 9 2) 6 3) 5 4) not defined
2. If $f(x) = ax + b$, where a and b are integers, $f(-1) = -5$ and $f(3) = 3$, then a and b are equal to
1) $a = -3, b = -1$ 2) $a = 2, b = -3$
3) $a = 0, b = 2$ 4) $a = 2, b = 3$

3. If $f(x) = \begin{cases} x^2 + 1, x \leq 0 \\ 2x - 1, 0 < x < 5 \\ 4x + 3, x \geq 5 \end{cases}$
then $\frac{f(-3) + f(2) + f(5)}{f(1)} =$
1) 28 2) 36 3) 26 4) 34
4. If $f = \{(-1, 3), (0, 2), (1, 1)\}$ then the range of $f^2 - 1$ is
1) $\{0, 8\}$ 2) $\{0, 3, 8\}$ 3) $\{0, 1, 3\}$ 4) $\{0, 2, 8\}$
5. $f: R \rightarrow R$ is a function defined by $f(x) = \frac{1}{2}$
then $f\left(-\frac{2}{5}\right) =$
1) 2 2) $-\frac{1}{2}$ 3) -2 4) $\frac{1}{2}$
6. If $f(x) = \sin\left(\frac{\pi}{3}[x] - x^2\right)$ then the value of $f\left(\sqrt{\frac{\pi}{3}}\right)$ is
1) 1 2) -1 3) 0 4) $-\frac{3}{4}$
7. The function $y = f(x)$ such that $f\left(x + \frac{1}{x}\right) = x^2 + \frac{1}{x^2}$
1) $2 - x^2$ 2) $x^2 - 2$ 3) $x^2 + 4$ 4) $4x^2 - 2$
8. If $f = \{(a, 1), (b, -2), (c, 3)\}$,
 $g = \{(a, -2), (b, 0), (c, 1)\}$
then $f^2 + g^2 =$
1) $\{(a, -1), (b, -2), (c, 4)\}$
2) $\{(a, 3), (b, -2), (c, 2)\}$
3) $\{(a, -4), (b, -4), (c, 9)\}$
4) $\{(a, 5), (b, 4), (c, 10)\}$
9. Which of the following relations are functions
 $f: \{(2, 1), (3, 1), (4, 2)\}$
 $g: \{(2, 2), (2, 4), (3, 3), (4, 4)\}$
 $h: \{(1, 2), (2, 3), (3, 4), (4, 5), (5, 6)\}$
1) f, g 2) g, h 3) h, f 4) f, g, h

EVEN AND ODD FUNCTIONS

10. If $f(x) = ax^5 + bx^3 + cx + d$ is odd then
 1) $a = 0$ 2) $b = 0$ 3) $c = 0$ 4) $d = 0$
11. A function whose graph is symmetrical about the y axis is given by

- 1) $f(x) = \sin\left[\log\left(x + \sqrt{x^2 + 1}\right)\right]$
 2) $f(x) = \frac{\sec^4 x + \cos^4 x}{x^3 + x^4 \cot x}$
 3) $f(x+y) = f(x) + f(y) \forall x, y \in R$
 4) $f(x) = x^2$

PERIODIC FUNCTIONS

12. The period of $\operatorname{sgn}(x - [x])$ is
 1) any real number 2) 0 3) 1
 4) non periodic function
13. The period of $x \cos x$ is
 1) 2π 2) π 3) $\frac{\pi}{2}$ 4) non periodic

DOMAIN OF THE FUNCTION

14. Domain of $f(x) = \frac{x^2 + 3x + 5}{x^2 - 5x + 4}$ is
 1) $R - \{1, 4\}$ 2) $\{1, 4\}$ 3) $(1, 4)$ 4) $(1, 4]$
15. Domain of $[x] + x$ is
 1) R 2) Z 3) $R - Z$ 4) Q
16. Domain of $|x - 1|$ is
 1) $[1, \infty)$ 2) R 3) $[0, \infty)$ 4) Z
17. The domain of $f(x) = \sqrt{-x^2}$ is
 1) $(0, \infty)$ 2) $(-\infty, 0)$ 3) $\{0\}$ 4) $(1, \infty)$
18. The domain of $f(x) = \frac{1}{|x| + x}$ is
 1) $(-\infty, 0)$ 2) $(0, \infty)$ 3) $(-\infty, 1)$ 4) $(-2, -1)$
19. The domain of $f(x) = \sqrt{1 - |x|}$ is
 1) $[-1, 1]$ 2) $(-1, 1)$ 3) $(0, 1)$ 4) R
20. The domain of $f(x) = \log\{(x-3)(6-x)\}$ is
 1) $(3, \infty)$ 2) $(3, 6)$ 3) $(0, \infty)$ 4) $(-\infty, \infty)$
21. The domain of $f(x) = \cot \frac{x}{3}$ is
 1) $(-\infty, \infty)$ 2) $R - \{n\pi : n \in Z\}$
 3) $R - \{3n\pi : n \in Z\}$ 4) $(0, \infty)$

22. The domain of $f(x) = \tan^{-1}(5x)$ is
 1) $(-\infty, \infty)$ 2) $(0, \infty)$ 3) $(-\infty, 0)$ 4) $\left(-\frac{1}{5}, \frac{1}{5}\right)$

23. The domain of $f(x) = \sqrt[3]{x} \cot x$ is
 1) R 2) $R - \{n\pi : n \in Z\}$
 3) $R - \left\{(2n+1)\frac{\pi}{2}, n \in Z\right\}$ 4) $(0, \infty)$

24. Domain of $f(x) = \frac{|x| - x}{2x}$ is
 1) R 2) $R - \{0\}$ 3) Z 4) N

25. Domain of $\frac{3^x}{x+1}$ is
 1) R 2) $R - \{-1\}$ 3) $(1, \infty)$ 4) $(-\infty, 1)$

RANGE OF THE FUNCTION

26. Range of $\frac{|x-4|}{x-4}$ is
 1) $R - \{4\}$ 2) R 3) $\{-1, 1\}$ 4) $R - \{-1, 1\}$
27. If x is positive, the values of
 $f(x) = -3 \cos \sqrt{3+x+x^2}$ lie in the interval
 1) $[-1, 3]$ 2) $[-3, 3]$ 3) $[0, 3]$ 4) $[-3, 0]$

28. The range of $f(x) = \frac{\sin \pi[x^2 - 1]}{x^4 + 1}$ is, where
 $[.]$ is greatest integer function
 1) R 2) $[-1, 1]$ 3) $\{0, 1\}$ 4) $\{0\}$

29. The range of $f(x) = |x-2| + |x-12|$ is
 1) $[2, \infty)$ 2) $(12, \infty)$ 3) $[10, \infty)$ 4) $[14, \infty)$
30. The range of $f(x) = 3x^2 + 7x + 10$ is
 1) $\left[\frac{70}{3}, \infty\right)$ 2) $\left[\frac{71}{12}, \infty\right)$ 3) $[0, \infty)$ 4) $\left(-\infty, \frac{70}{3}\right)$

31. The range of the function
 $f(x) = \frac{2+x}{2-x}, x \neq 2$ is [AIE-2002]

- 1) R 2) $R - \{-1\}$ 3) $R - \{1\}$ 4) $R - \{2\}$
32. The range of $f(x) = x - [x]$ is
 1) $x = \{1, 2, 3, \dots\}$ 2) $x \geq 0$
 3) $x < 1$ 4) $0 \leq x < 1$

33. Range of $\sqrt{9-x^2}$ is
 1) $[0, 3]$ 2) $[-3, 3]$ 3) $[-3, 0]$ 4) R

34. Range of $f(x) = e^x$ is
 1) $(0, \infty)$ 2) $[0, \infty)$ 3) $(-\infty, \infty)$ 4) $[e, \infty)$

TYPES OF FUNCTIONS

35. If $f: \mathbb{R} \rightarrow \mathbb{S}$, defined by $f(x) = \sin x - \sqrt{3} \cos x + 1$, is onto, then the interval of 'S' is [AIE- 2004]
 1) $[0,3]$ 2) $[-1,3]$ 3) $[0,1]$ 4) $[-1,1]$
36. If $f: [1, \infty) \rightarrow B$ defined by $f(x) = x^2 - 2x + 6$ is a surjection then $B =$
 1) $[1, \infty)$ 2) $[5, \infty)$ 3) $[6, \infty)$ 4) $[2, \infty)$
37. $f: (-\infty, \infty) \rightarrow (-\infty, \infty)$ is defined by $f(x) = ax + b, a, b \in \mathbb{R} (a \neq 0)$ then f is
 1) injective but not surjective
 2) surjective but not injective 3) bijective
 4) neither injective nor surjective
38. $f: \mathbb{Z} \rightarrow \mathbb{Z}$ and $f(x) = x^2$ then f is (EAM-95)
 1) bijection 2) injection
 3) surjection 4) not bijection
39. The function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \sin x$ is [AIE-2002]
 1) Neither one one nor onto 2) onto
 3) one-one 4) many one
40. $f: \mathbb{Z} \rightarrow \mathbb{Z}$ defined as $f(x) = [x]$ then f is
 1) not a function 2) many-to-one function
 3) into function 4) identity function
41. $f: \mathbb{Q} \rightarrow \mathbb{Q}$ is defined by $f(x) = 15x + 7$ is
 1) injective only 2) surjective only
 3) bijective 4) neither injective nor surjective
42. $f: (0, \infty) \rightarrow [0, \infty)$ defined by $f(x) = x^2$ is
 1) one-one but not onto 2) onto but not one-one
 3) bijective 4) neither one-one nor onto
43. Let $f: [0, \infty) \rightarrow [0, 2]$ be defined by $f(x) = \frac{2x}{1+x}$ then f is
 1) one-one but not onto 2) onto but not one-one
 3) both one-one and onto
 4) neither one-one nor onto
- Number of functions:**
44. The total number of functions from A to itself is 256, then $n(A) =$
 1) 2 2) 3 3) 4 4) 5
45. If $B = \{1, 2, 3\}$ and $A = \{4, 5, 6, 7, 8\}$ then the number of surjections from A to B is
 1) 81 2) 64 3) 48 4) 150

46. The number of one-one functions that can be defined from $A = \{1, 2, 3\}$ to $B = \{a, e, i, o, u\}$ is
 1) 3^5 2) 5^3 3) 5P_3 4) $5!$
47. The number of possible many to one functions from $A = \{6, 36\}$ to $B = \{1, 2, 3, 4, 5\}$ is
 1) 32 2) 25 3) 5 4) 20
48. If $n(A) = 4$ and $n(B) = 6$, then the number of surjections from A to B is
 1) 4^6 2) 6^4 3) 0 4) 24
49. The number of bijections from the set A to itself when A contains 106 elements is
 1) 106 2) 106^2 3) $106!$ 4) 2^{106}
50. The number of non-surjective mappings that can be defined from $A = \{1, 4, 9, 16\}$ to $B = \{2, 8, 16, 32, 64\}$ is
 1) 1024 2) 20 3) 505 4) 625

COMPOSITE FUNCTIONS

51. If $f(x) = 2x + 1$ and $g(x) = x^2 + 1$ then $g(f(f(2))) =$
 1) 112 2) 122 3) 12 4) 124
52. If $f(x) = [x], g(x) = x - [x]$ then which of the following functions is a zero function
 1) $(f+g)(x)$ 2) $(fg)(x)$
 3) $(f-g)(x)$ 4) $(fog)(x)$
53. Let $f(x) = \frac{Kx}{x+1} (x \neq -1)$ then the value of K for which $(f \circ f)(x) = x$ is
 1) 1 2) -1 3) 2 4) $\sqrt{2}$
54. If $f(x) = \frac{x}{\sqrt{1-x^2}}, g(x) = \frac{x}{\sqrt{1+x^2}}$ then $(fog)(x) =$
 1) x 2) $\frac{x}{\sqrt{1+x^2}}$ 3) $\sqrt{1+x^2}$ 4) $2x$
55. If $f: [-6, 6] \rightarrow \mathbb{R}$ is defined by $f(x) = x^2 - 3$ for $x \in \mathbb{R}$ then (EAM-08)
 $(f \circ f \circ f)(-1) + (f \circ f \circ f)(0) + (f \circ f \circ f)(1) =$
 1) $f(4\sqrt{2})$ 2) $f(3\sqrt{2})$ 3) $f(2\sqrt{2})$ 4) $f(\sqrt{2})$
56. If $f(x) = (1-x)^{1/2}$ and $g(x) = \ln(x)$ then the domain of $(g \circ f)(x)$ is
 1) $(-\infty, 2)$ 2) $(-1, 1)$ 3) $(-\infty, 1]$ 4) $(-\infty, 1)$

INVERSE OF A FUNCTION

57. If $f: R^+ \rightarrow R$ such that $f(x) = \log_5 x$
then $f^{-1}(x) =$ (EAM-97)
1) $\log_x 10$ 2) 5^x 3) 3^{-x} 4) $3^{1/x}$
58. $f: R \rightarrow R$ is a function defined by
 $f(x) = 10x - 7$. If $g = f^{-1}$ then $g(x) =$
1) $\frac{1}{10x-7}$ 2) $\frac{1}{10x+7}$ 3) $\frac{x+7}{10}$ 4) $\frac{x-7}{10}$
59. $f: [0, \infty) \rightarrow [4, \infty)$ is defined by
 $f(x) = x^2 + 4$ then $f^{-1}(13) =$
1) 3 2) 2 3) 1 4) 4
60. If the function $f: R \rightarrow R$ be such that
 $f(x) = x - [x]$, where $[x]$ denotes the
greatest integer less than or equal to x then
 $f^{-1}(x)$ is
1) $\frac{1}{x-[x]}$ 2) $[x] - x$
3) not defined 4) $x - [x]$

LEVEL-I(H.W.)-KEY

- 01) 4 02) 2 03) 2 04) 2 05) 4 06) 3
07) 2 08) 4 09) 3 10) 4 11) 4 12) 3
13) 4 14) 1 15) 1 16) 2 17) 3 18) 2
19) 1 20) 2 21) 3 22) 1 23) 2 24) 2
25) 2 26) 3 27) 2 28) 4 29) 3 30) 2
31) 2 32) 4 33) 1 34) 1 35) 2 36) 2
37) 3 38) 4 39) 1 40) 4 41) 3 42) 1
43) 1 44) 3 45) 4 46) 3 47) 3 48) 3
49) 3 50) 4 51) 2 52) 4 53) 2 54) 1
55) 1 56) 4 57) 2 58) 3 59) 1 60) 3

LEVEL-I(H.W.)-HINTS

- $f(2)$ is not defined from the definition.
- Solve the equations
- Substitute the values
- Range of $f^2 - 1 = \{3^2 - 1, 2^2 - 1, 1^2 - 1\}$
- Constant function
- $\left[\sqrt{\pi/3}\right] = 1 \Rightarrow f(x) = \sin\left(\frac{\pi}{3} - x^2\right)$
- $f\left(x + \frac{1}{x}\right) = \left(x + \frac{1}{x}\right)^2 - 2$
- $f^2 + g^2 = \{(a, 1+4), (b, 4+0), (c, 9+1)\}$

- Concept
- $f(-x) = -f(x)$
- A function whose graph is symmetrical about the y-axis must be even since $\sin x$ and $\log\left(x + \sqrt{x^2 + 1}\right)$ are odd function.
Therefore $\sin\left(\log\left(x + \sqrt{x^2 + 1}\right)\right)$ must be odd
Also, $\frac{\sec^4 + \operatorname{cosec}^4 x}{x^3 + x^4 \cot x}$ is an odd function
Now, let $f(x) + y = f(x) + f(y) \forall x, y \in R$
 $\therefore f(0+0) = f(0) + f(0) \therefore f(0) = 0$
 $f(x-x) = f(x) + f(-x)$ or $0 = f(x) + f(-1)$
i.e., $f(-x) = -f(x) \therefore f(x)$ is odd.
- $\{x\} > 0$ if $x \notin Z$, $\{x\} = 0$ if $x \in Z$
- no value of T exist such that $f(x+T) = f(x)$
- denominator not equal to zero
- $[x]$, x is defined for all values
- draw graph.
- $\sqrt{f(x)}$ defined $f(x) \geq 0$.
- $|x| + x \neq 0$
- $1 - |x| \geq 0$
- $(x-3)(6-x) > 0$
- Domain of $\cot x = R \setminus \{n\pi, n \in Z\}$
- $5x \in R$
- Domain of $\cot x = R - \{n\pi, n \in Z\}$
- Denominator not equal to zero
- Denominator $\neq 0$
- $x \geq 4$ then $f(x) = 1$; $x < 4$ then $f(x) = -1$
- Range of $\cos x = [-1, 1]$
- $[x^2 + 1]$ is an integer $\Rightarrow$
 $\sin n\pi = 0 \forall x \in R \Rightarrow f(x) = 0 \forall x \in R$
- If $2 \leq x < 12$ then range of $f = \{10\}$
if $x \geq 12$ then the range of $f = [10, \infty)$
if $x \geq 2$ then the range of $f = [10, \infty)$
- Range of $ax^2 + bx + c$ ($a > 0$) is $\left[\frac{4ac - b^2}{4a}, \infty\right)$

31. $y = \frac{2+x}{2-x} \Rightarrow 2y - yx = 2 + x \Rightarrow x(y+1) = 2y-2$
 $x = \frac{2y-2}{y+1} \Rightarrow f^{-1}(x) = \frac{2x-2}{x+1}$
 $\therefore \text{Range } f = \text{domain } f^{-1} = \mathbb{R} - \{-1\}$
32. $x - [x] \in [0, 1)$
33. substitute domain values
34. Use graph
35. $f: \mathbb{R} \rightarrow S$ is onto $\Rightarrow S = \text{Range}$
 $f = [1 - \sqrt{1+3}, 1 + \sqrt{1+3}] = [-1, 3]$
36. $x^2 - 2x + 1 + 5 = (x-1)^2 + 5 \geq 5$
37. $f(x_1) = f(x_2) \Rightarrow x_1 = x_2 \Rightarrow f$ is one-one
 let $f(x) = y \Rightarrow ax + b = y \Rightarrow x = \frac{y-b}{a} \in \mathbb{R}$
 $\forall y \in \mathbb{R} \Rightarrow f$ is onto
38. $f(-1) = f(1) = 1 \Rightarrow f$ is not one-one
 $x^2 > 0 \forall x \in \mathbb{R} \Rightarrow \text{Range of } f \neq \text{codomain}$
39. $\sin x$ is a periodic function with period $2\pi \Rightarrow \sin x$ is a many one function.
40. $f(x) = [x] = x \forall x \in \mathbb{R}$, f is identity function
41. $f(x_1) = f(x_2) \Rightarrow x_1 = x_2 \Rightarrow f$ is one-one
 Let $f(x) = y \Rightarrow 15x + 7 = y \Rightarrow x = \frac{y-7}{15} \in \mathbb{Q}$
 $\forall y \in \mathbb{Q} \Rightarrow f$ is onto
42. $f(x_1) = f(x_2) \Rightarrow x_1^2 = x_2^2 \Rightarrow x_1 = x_2 \Rightarrow f$ is one-one
 Range of $f = (0, \infty)$ not equal to codomain
43. $f(x_1) = f(x_2); \frac{2x_1}{1+x_1} = \frac{2x_2}{1+x_2} \Rightarrow x_1 = x_2$
 $\therefore f$ is one-one, $\text{Range} \subset \text{Co-Domain}$
 $\therefore f$ is not onto
44. No. of function from A to A = $n(A)^{n(A)} = 256$
45. The number of surjections from A to B is
 $n^r - {}^nC_1(n-1)^r + {}^nC_2(n-2)^r - {}^nC_3(n-3)^r + \dots + (-1)^{n-1} \cdot {}^nC_{n-1}$ where $n(A) = r$ & $n(B) = n$
46. No. of one-one functions from A to B = ${}^{n(B)}P_{n(A)}$
 $(n(B) \geq n(A))$
47. No. of many-one functions from A to B
 $= n(B)^{n(A)} - {}^{n(B)}P_{n(A)}$

48. If $n(A) < n(B)$ then the no. of surjections from A to B = 0
49. $n(A)! = 106!$
50. $5^4 - 0 = 625$
51. $g(f(f(2))) = g(f(5)) = g(11) = 121 + 1 = 122$
 $(f \circ g)(x) = f[g(x)] = f[x - [x]]$
 $= f(\{x\}) = [\{x\}] = 0, \sin 0 \leq \{x\} < 1$
52. $\text{fof}(x) = x \Rightarrow f(f(x)) = x$
53. $f(g(x)) = \frac{g(x)}{\sqrt{1-(g(x))^2}} = \frac{\frac{x}{\sqrt{1+x^2}}}{\sqrt{1-\frac{x^2}{1+x^2}}} = x$
54. Substitute the values
55. $x < 1$
56. let $f^{-1}(x) = y \Rightarrow x = f(y) = \log_5 y$
 $\Rightarrow y = f^{-1}(x) = 5^x$
57. Let $f^{-1}(x) = y \Rightarrow x = f(y) = 10y - 7$
 $\Rightarrow y = f^{-1}(x) = \frac{x+7}{10} = g(x)$
58. Let $f^{-1}(x) = y \Rightarrow x = f(y) = y + \frac{1}{y}$
 $\Rightarrow y^2 - xy + 1 = 0$
59. $x - [x] = 0$ for all integral values of x . Therefore, the function is many one-and, therefore, not defined.



LEVEL - II (C.W)

REAL VALUED FUNCTIONS

1. Let $g(x)$ be a function defined on $[-1, 1]$. If the area of the equilateral triangle with two of its vertices at $(0, 0)$ and $(x, g(x))$ is $\sqrt{3}/4$, then the function $g(x)$ is
 1) $g(x) = \pm\sqrt{1-x^2}$ 2) $g(x) = \sqrt{1-x^2}$
 3) $g(x) = -\sqrt{1+x^2}$ 4) $g(x) = \sqrt{1+x^2}$
2. If $f: \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = x - [x] - \frac{1}{2}$ for $x \in \mathbb{R}$, where $[x]$ is the greatest integer not exceeding x , then $\left\{x \in \mathbb{R}: f(x) = \frac{1}{2}\right\} =$
 1) $\mathbb{Z}$ 2) $\mathbb{N}$ 3) $\emptyset$ 4) $\mathbb{R}$

3. Suppose $f : [2, 2] \rightarrow R$ is defined by

$$f(x) = \begin{cases} -1 & \text{for } -2 \leq x \leq 0 \\ x-1 & \text{for } 0 \leq x \leq 2 \end{cases}$$

then the $\{x \in (-2, 2) : x \leq 0 \text{ and } f(|x|) = x\} =$

- 1) $\{-1\}$ 2) $\{0\}$ 3) $\{-1/2\}$ 4) $\emptyset$

4. If $|\sin x + \cos x| = |\sin x| + |\cos x|$, then x lies in

- 1) 1st quadrant only 2) 1st and 3rd quadrant only
3) 2nd and 4th quadrant only 4) 3rd and 4th quadrant only

EVEN AND ODD FUNCTIONS

5. Let $f(x) = \begin{cases} 4 & x < -1 \\ -4x & -1 \leq x \leq 0 \end{cases}$

If $f(x)$ is an even function on R then the definition of $f(x)$ on $(0, \infty)$ is

1) $f(x) = \begin{cases} 4x & 0 < x \leq 1 \\ 4 & x > 1 \end{cases}$ 2) $f(x) = \begin{cases} 4x & 0 < x \leq 1 \\ -4 & x > 1 \end{cases}$

3) $f(x) = \begin{cases} 4 & 0 < x \leq 1 \\ 4x & x > 1 \end{cases}$ 4) $f(x) = \begin{cases} 4 & x < -1 \\ -4x & -1 \leq x \leq 0 \end{cases}$

6. If $f(x) = \begin{cases} x^2 \sin \frac{\pi x}{2} & |x| < 1 \\ x|x| & |x| \geq 1 \end{cases}$ then $f(x)$ is

- 1) an even function 2) an odd function
3) a periodic function 4) neither odd nor even

7. $f(x) = \left[\frac{\cos x}{\frac{2x}{\pi}} + \frac{1}{2} \right]$, where x is not an

integral multiple of π and $[\cdot]$ denotes the greatest integer function is

- 1) an odd function 2) even function
3) neither odd nor even 4) both even and odd

PERIODIC FUNCTIONS

8. Which of the following function is not periodic

- 1) $\frac{2^x}{2^{[x]}}$ 2) $\sin^{-1}(\{x\})$
3) $\sin^{-1}(\sqrt{\cos x})$ 4) $\sin^{-1}(\cos(x^2))$

9. Let $f(x) = nx + n - [nx + n] + \tan \frac{\pi x}{2}$, where

$[x]$ is the greatest integer $\leq x$ and $n \in N$. It is

- 1) a periodic function of period 1
2) a periodic function of period 4
3) not periodic
4) a periodic function of period 2

10. Let $f(x) = x(2-x)$, $0 \leq x \leq 2$. If the

definition of f is extended over the set

$R - [0, 2]$ by $f(x+2) = f(x)$ then f is a

- 1) periodic function of period 1
2) non periodic function
3) periodic function of period 2
4) periodic function of period $1/2$

11. If f is periodic, g is polynomial function and

$f(g(x))$ is periodic and $g(2) = 3, g(4) = 7$

then $g(6)$ is

- 1) 13 2) 15 3) 11 4) 21

DOMAIN OF THE FUNCTION

12. The domain of the function

$f(x) = \frac{\sin^{-1}(x-3)}{\sqrt{9-x^2}}$ is [AIE-2004]

- 1) $[2, 3]$ 2) $[1, 2]$ 3) $[1, 2]$ 4) $[2, 3]$

13. The domain of $f(x) = \sin^{-1} \left\{ \log_3 \left(\frac{x^2}{3} \right) \right\}$ is

- 1) $(-\infty, 3]$ 2) $[3, \infty)$
3) $[-3, -1] \cup [1, 3]$ 4) $(-9, -1) \cup (1, 9)$

14. The domain of $f(x) = \log_x(9-x^2)$ is

- 1) $(-3, 3)$ 2) $(0, \infty)$
3) $(0, 1) \cup (1, \infty)$ 4) $(0, 1) \cup (1, 3)$

15. The domain of $f(x) = \frac{3}{4-x^2} + \log_{10}(x^3 - x)$ is [AIE-2003]

- 1) $(1, 2)$ 2) $(-1, 0) \cup (1, 2)$
3) $(1, 2) \cup (2, \infty)$ 4) $(-1, 0) \cup (1, 2) \cup (2, \infty)$

16. The domain of $f(x) = \sqrt{1 - \sqrt{1 - \sqrt{1 - x^2}}}$ is

- 1) $[0, 1]$ 2) $[-1, 1]$ 3) $(-\infty, \infty)$ 4) $(-1, 1)$

17. The domain of $f(x) = \cos(\log x)$ is

- 1) $(-\infty, \infty)$ 2) $(-1, 1)$ 3) $(0, \infty)$ 4) $(1, \infty)$

18. The domain of $f(x) = \cos^{-1}\left(\frac{2}{2 + \sin x}\right)$ contained in $[0, 2\pi]$ is

- 1) $\left[0, \frac{\pi}{2}\right]$ 2) $\left[\frac{\pi}{2}, \pi\right]$ 3) $[0, \pi]$ 4) $\left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$

19. $\{x \in R : [x - |x|] = 5\} =$ (EAM-05)

- 1) R is the set of real numbers
2) ϕ , the null set
3) $\{x \in R : x < 0\}$ 4) $\{x \in R : x \geq 0\}$

20. The domain of the function defined by

$$f(x) = {}^{(7-x)}P_{(x-3)} \text{ is}$$

- 1) $\{3, 7\}$ 2) $\{3, 4, 5, 6, 7\}$ 3) $\{3, 4, 5\}$ 4) $\{1, 2, 3, 4\}$

21. $f : N \rightarrow N$ is defined as

$$f(n) = \begin{cases} 2, n = 3k, & k \in Z \\ 10 - n, n = 3k + 1, & k \in Z \\ 0, n = 3k + 2, & k \in Z \end{cases}$$

then $\{n \in N : f(n) > 2\} =$ (EAM-04)

- 1) $\{3, 6, 4\}$ 2) $\{1, 4, 7\}$ 3) $\{4, 7\}$ 4) $\{7\}$

22. The domain of the function

$$f(x) = \frac{1}{\sqrt{\{\sin x\} + \{\sin(\pi + x)\}}} \text{ where } \{\cdot\}$$

denotes the fractional part, is

1. $[0, \pi]$ 2. $(2n+1)\frac{\pi}{2}, n \in Z$
3. $(0, \pi)$ 4. $R - \left\{\frac{n\pi}{2}, n \in Z\right\}$

RANGE OF THE FUNCTION

23. If $\alpha \in (0, \frac{\pi}{2})$, then $\sqrt{x^2 + x} + \frac{\tan^2 \alpha}{\sqrt{x^2 + x}}$

is always greater than or equal to $(x \neq 0, -1)$

- 1) 2 2) 1 3) $2 \tan \alpha$ 4) $2 \sec^2 \alpha$

24. The range of $f(x) = \sin^2 x + \cos^4 x$ is

- 1) $\left[\frac{1}{2}, 1\right]$ 2) $\left[\frac{3}{4}, 1\right]$ 3) $[0, 1]$ 4) $\left[0, \frac{1}{4}\right]$

25. The range of $f(x) = \sin^{-1}\left[\frac{1}{2} + x^2\right]$ is $[\cdot]$ denotes greatest integer function)

- 1) $\left\{-\frac{\pi}{2}, 0, \frac{\pi}{2}\right\}$ 2) $\left\{0, \frac{\pi}{2}\right\}$ 3) $\left\{\frac{\pi}{2}\right\}$ 4) $\{0, \pi\}$

26. If $f : R \rightarrow R$ and $g : R \rightarrow R$ defined by

$$f(x) = |x| \text{ and } g(x) = [x - 3] \text{ for}$$

$x \in R$ $[\cdot]$ is denotes greatest integer function

then $\left\{g(f(x)) : \frac{-8}{5} < x < \frac{8}{5}\right\} =$ (EAM-08)

- 1) $\{0, 1\}$ 2) $\{1, 2\}$ 3) $\{-3, -2\}$ 4) $\{2, 3\}$

27. The range of $x^2 + 4y^2 + 9z^2 - 6yz - 3xz - 2xy$ is

- 1) ϕ 2) R 3) $[0, \infty)$ 4) $(-\infty, 0)$

28. The maximum possible domain and the

corresponding range of $f(x) = (-1)^x$ are

- 1) $D_f = R, R_f = [-1, 1]$ 2) $D_f = Z, R_f = \{1, -1\}$
3) $D_f = Z, R_f = [-1, 1]$ 4) $D_f = R, R_f = \{-1, 1\}$

29. The range of

$$f(x) = \sin^{-1} x + \cos^{-1} x + \tan^{-1} x \text{ is}$$

- 1) $(0, \pi)$ 2) $\left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$ 3) $\left[\frac{-\pi}{4}, \frac{\pi}{4}\right]$ 4) $\left[0, \frac{3\pi}{4}\right]$

TYPES OF FUNCTIONS

30. Let $A = [-1, 1] = B$ then which of the following function from A to B is bijective function

- 1) $f(x) = \frac{x}{2}$ 2) $g(x) = |x|$
3) $h(x) = x^2$ 4) $k(x) = \sin \frac{\pi x}{2}$

31. If $f : R \rightarrow C$ is defined by

$$f(x) = e^{2ix} \text{ for } x \in R \text{ then, } f \text{ is (Where } C$$

denotes the set of all Complex numbers)

- 1) One-one 2) Onto
3) One-one and Onto
4) neither one-one nor Onto

32. A function $f : N \rightarrow Z$ defined by

$$f(n) = \begin{cases} \frac{n-1}{2}, & \text{when 'n' is odd} \\ \frac{-n}{2}, & \text{when 'n' is even} \end{cases}, \text{ is}$$

- 1) one-one but not onto 2) onto but not one-one
3) one-one onto 4) neither one-one nor onto

33. **M** is the set of all 2×2 real matrices.

$f: M \rightarrow R$ is defined by $f(A) = \det A$ for all A in **M** then **f** is

- 1) one-one but not onto 2) onto but not one-one
3) neither one-one nor onto 4) bijective

34. Let $f: R - \{n\} \rightarrow R$ be a function defined

by $f(x) = \frac{x-m}{x-n}$ such that $m \neq n$ then

- 1) f is one one into function
2) f is one one onto function
3) f is many one into function
4) f is many one onto function

35. $f(x) = \begin{cases} 0, & \text{if } x \text{ is rational} \\ x, & \text{if } x \text{ is irrational} \end{cases}$

$g(x) = \begin{cases} 0, & \text{if } x \text{ is irrational} \\ x, & \text{if } x \text{ is rational} \end{cases}$ Then $f - g$ is

- 1) one-one and into 2) neither one-one nor onto
3) many one and onto 4) one-one and onto

COMPOSITE FUNCTIONS

36. If $f(x)$ and $g(x)$ are two functions with

$g(x) = x - \frac{1}{x}$ and $f \circ g(x) = x^3 - \frac{1}{x^3}$, then $f(x) =$

- 1) $x^3 + 3x$ 2) $x^2 - \frac{1}{x^2}$ 3) $1 + \frac{1}{x^2}$ 4) $3x^2 + \frac{3}{x^4}$

37. Let $f(x) = ax + b$ and

$g(x) = cx + d$, $a \neq 0, c \neq 0$. Assume

$a = 1, b = 2$. If $(f \circ g)(x) = (g \circ f)(x)$ for all x , what can you say about c and d

- 1) c and d both arbitrary 2) $c = 1, d$ arbitrary
3) c arbitrary, $d = 1$ 4) $c = 1, d = 1$

38. If $f(x) = \sin^2 x$ and the composite functions

$g\{f(x)\} = |\sin x|$, then the function $g(x) =$

- 1) $\sqrt{x-1}$ 2) $\sqrt{x}$ 3) $\sqrt{x+1}$ 4) $-\sqrt{x}$

39. If $f: R \rightarrow R$ and $g: R \rightarrow R$ are given by

$f(x) = |x|$ and $g(x) = [x]$ for each $x \in R$,

then $\{x \in R : g(f(x)) \leq f(g(x))\} =$

- 1) $Z \cup (-\infty, 0)$ 2) $(-\infty, 0)$ 3) Z 4) R

40. Let $g: R \rightarrow R$ be given by $g(x) = 3 + 4x$. If

$g^n(x) = g \circ g \circ \dots \circ g(x)$, and $g^n(x) = A + Bx$ then **A** and **B** are

- 1) $2^{n+1} - 1, 2^{n+1}$ 2) $4^n - 1, 4^n$
3) $3^n, 3^n + 1$ 4) $5^n - 1, 5^n$

INVERTIBILITY & INVERSE OF A FUNCTION

41. Let '**f**' be an injective function with domain $\{x, y, z\}$ and range $\{1, 2, 3\}$ such that exactly one of the following statements is correct and the remaining are false $f(x) = 1$,

$f(y) \neq 1, f(z) \neq 2$ the value of $f^{-1}(1)$ is

- 1) x 2) y 3) z 4) x or z

42. If the function $f: [1, \infty) \rightarrow [1, \infty)$ is defined

by $f(x) = 2^{x(x-1)}$ then $f^{-1}(x)$ is

- 1) $\left(\frac{1}{2}\right)^{x(x-1)}$ 2) $\frac{1}{2}(1 + \sqrt{1 + 4 \log_2 x})$
3) $\frac{1}{2}(1 - \sqrt{1 + 4 \log_2 x})$ 4) $\frac{1}{2}(1 \pm \sqrt{1 + 4 \log_2 x})$

FUNCTIONAL TRANSFORMATIONS

43. If $f(x)$ is a polynomial in $x (> 0)$ satisfying the equation

$f(x) + f(1/x) = f(x) \cdot f(1/x)$ and

$f(2) = -7$, then $f(3) =$

- 1) -26 2) -27 3) -28 4) -29

44. A function $f: R \rightarrow R$ satisfy the equation

$f(x)f(y) - f(xy) = x + y$ for all $x, y \in R$ and $f(y) > 0$, then

- 1) $f(x) = x + \frac{1}{2}$ 2) $f(x) = \frac{x}{2} + 1$
3) $f(x) = \frac{x}{2} - 1$ 4) $f(x) = x + 1$

45. **f** is a function defined as

$\sum_{k=1}^n f(a+k) = 16(2^n - 1)$ and $f(x+y) = f(x) \cdot f(y)$

and $f(1) = 2$ then integral value of **a**

- 1) 3 2) 0 3) 2 4) 1

46. A real valued function **f(x)** satisfies the functional equation

$f(x-y) = f(x)f(y) - f(a-x)f(a+y)$

where **a** is a given constant and $f(0) = 1$,

$f(2a-x)$ is equal to [AIE-2004]

- 1) $f(-x)$ 2) $f(a) + f(a-x)$
3) $f(x)$ 4) $-f(x)$

LEVEL-II(C.W)-KEY

01) 2	02) 3	03) 3	04) 2	05) 1	06) 2
07) 1	08) 4	09) 4	10) 3	11) 3	12) 4
13) 3	14) 4	15) 4	16) 2	17) 3	18) 3
19) 2	20) 3	21) 2	22) 4	23) 3	24) 2
25) 2	26) 3	27) 3	28) 2	29) 2	30) 4
31) 4	32) 3	33) 2	34) 1	35) 4	36) 1
37) 2	38) 2	39) 4	40) 2	41) 2	42) 2
43) 1	44) 4	45) 1	46) 4		

LEVEL-II(C.W)-HINTS

- Side of the triangle with vertices $(0, 0)$ and $(x, g(x))$ is a $\sqrt{x^2 + (g(x))^2}$. Area of equilateral triangle, whose side a is $\frac{\sqrt{3}}{4} a^2$.
 $\therefore \frac{\sqrt{3}}{4} [x^2 + g(x)^2] = \frac{\sqrt{3}}{4} \Rightarrow x^2 + g(x)^2 = 1$
 $\Rightarrow g(x) = \pm \sqrt{1 - x^2}$. Thus $g(x) = \sqrt{1 - x^2}$
- $0 \leq x - [x] < 1 \Rightarrow -\frac{1}{2} \leq x - [x] - \frac{1}{2} < \frac{1}{2}$
 $\Rightarrow -\frac{1}{2} \leq f(x) < \frac{1}{2} \Rightarrow f(x) \neq \frac{1}{2} \forall x$
 $\Rightarrow \left\{ x \in \mathbb{R} : f(x) = \frac{1}{2} \right\} = \phi$
- By verification, $f\left(\left|-\frac{1}{2}\right|\right) = f\left(\frac{1}{2}\right) = \frac{1}{2} - 1 = -\frac{1}{2}$
hence $f(|x|) = x$.
- The given relation holds only when $\sin x$ and $\cos x$ have same sign or atleast one of them is zero.
hence $x \in [0, \pi/2] \cup [\pi, 3\pi/2]$
- If $f(x) = \begin{cases} 4, & x < -1 \\ -4x & -1 \leq x \leq 0 \end{cases}$ is even,
 $f(x) = f(-x)$
 $f(x) = \begin{cases} 4, & -x < -1 \\ -4x & -1 \leq -x \leq 0 \end{cases} = \begin{cases} 4x, & 0 \leq x \leq 1 \\ 4, & x > 1 \end{cases}$
- Here, $f(x) = x|x|, x \leq -1$

$$x^2 \sin\left(\frac{\pi x}{2}\right), -1 < x < 1, x|x|, x \geq 1.$$

Let $k \geq 0$. Then

$$\text{and } f(1+k) = (1+k)|1+k| = (1+k)^2.$$

$$\text{Therefore, } f(1+k) = -f(-1+k)$$

$$f(-1+k) = (-1+k)^2 \sin \frac{\pi}{2}(-1+k) =$$

$$-(1-k)^2 \sin \frac{\pi}{2}(1-k) = -f(1-k) = -f|-(1+k)|$$

$\therefore f(x) = -f(-x)$ for all x . Also, none of the pieces of definition are periodic.

$$7. \quad f(-x) = \frac{\cos(-x)}{\left[-\frac{2x}{\pi}\right] + \frac{1}{2}} = \frac{\cos x}{-1 - \left[\frac{2x}{\pi}\right] + \frac{1}{2}}$$

(as x is not an integral multiple of π)

$$\Rightarrow f(-x) = -\frac{\cos x}{\left[\frac{2x}{\pi}\right] + \frac{1}{2}} = -f(x)$$

$\Rightarrow f(x)$ is an odd function

- Period of $\frac{2^x}{2^{[x]}}$ is 1. period of $\sin^{-1}\{x\}$ is 1. period of $\sin^{-1}(\sqrt{\cos x})$ is 2π whereas $\sin^{-1}(\cos x^2)$ is non-periodic.

- $nx + n - [nx + n]$ has the period $1/n$ and $\tan \frac{\pi x}{2}$

has the period $\frac{\pi}{\pi/2}$ i.e, LCM of $1/n, 2$ is 2.

- Conceptual

- From the given data $g(x)$ must be linear function. Hence, $g(x) = ax + b$

Also $g(2) = 2a + b = 3$ and $g(4) = 4a + b = 7$ solving, $a = 2$ and $b = -1$

Hence, $g(x) = 2x - 1$ Then, $g(6) = 11$

- $f(x)$ is defined $\Rightarrow -1 \leq x - 3 \leq 1$ and

$$9 - x^2 > 0 \Rightarrow 2 \leq x \leq 4 \text{ and } -3 < x < 3.$$

$$\therefore \text{domain} = [2, 3]$$

$$13. -1 \leq \log_3 \left(\frac{x^2}{3} \right) \leq 1 \Rightarrow 1 \leq x^2 \leq 9$$

$$14. 9 - x^2 > 0, x > 0 \text{ and } x \neq 1$$

$$15. \frac{3}{4-x^2} + \log_{10}(x^3 - x) \text{ is defined}$$

$$\Rightarrow 4 - x^2 \neq 0, x^3 - x > 0 \Rightarrow x \neq \pm 2,$$

$$(x+1)x(x-1) > 0$$

$$\therefore \text{domain} = (-1, 0) \cup (1, 2) \cup (2, \infty)$$

$$16. 1 - x^2 \geq 0$$

$$17. x > 0$$

$$18. -1 \leq \frac{2}{2+\sin x} \leq 1$$

$$19. [x - [x]] \neq 5 \forall x \in \mathbb{R}$$

$$20. x \in \mathbb{Z}, 7 - x \geq x - 3, 7 - x > 0, x - 3 \geq 0$$

$$21. \text{Verification}$$

$$22. f(x) = \frac{1}{\sqrt{\{\sin x\} + \{\sin(\pi + x)\}}}$$

$$= \frac{1}{\sqrt{\{\sin x\} + \{-\sin x\}}}$$

$$\text{Now } \{\sin x\} + \{-\sin x\} = \begin{cases} 0 & \sin x \text{ is an integer} \\ 1 & \sin x \text{ is not an integer} \end{cases}$$

$$\text{For } f(x) \text{ to get defined } \{\sin x\} + \{-\sin x\} \neq 0$$

$$\Rightarrow \sin x \neq \text{integer} \Rightarrow \sin x \neq \pm 1, 0 \Rightarrow x \neq \frac{n\pi}{2}, n \in \mathbb{I}$$

$$\text{Hence, the domain is } \mathbb{R} - \left\{ \frac{n\pi}{2} / n \in \mathbb{I} \right\}$$

$$23. A.M. \geq G.M.$$

$$24. \sin^2 x + \cos^4 x = \sin^2 x + \cos^2 x (1 - \sin^2 x)$$

$$= 1 - \frac{1}{4} \sin^2 2x \in [3/4, 1]$$

$$25. 0 \leq \left[\frac{1}{2} + x^2 \right] \leq 1 \Rightarrow \left[\frac{1}{2} + x^2 \right] = 0, 1$$

$$26. -\frac{8}{5} < x < \frac{8}{5} \Rightarrow 0 \leq |x| < \frac{8}{5} \Rightarrow 0 \leq f(x) < \frac{8}{5}$$

$$27. x^2 + (2y)^2 + (3z)^2 - (2y)(3z) - (x)(3z) - x(2y) \geq 0$$

$$28. f(x) = (-1)^x \text{ is defined when } x \text{ is an integer}$$

$$\therefore \text{Domain} = \mathbb{Z}, \text{range} = \{-1, 1\}$$

$$29. \text{Domain } [-1, 1]$$

$$-1 \leq x \leq 1 \Rightarrow \frac{-\pi}{4} \leq \tan^{-1} x \leq \frac{\pi}{4}$$

$$\frac{\pi}{2} - \frac{\pi}{4} \leq f(x) \leq \frac{\pi}{2} + \frac{\pi}{4} \Rightarrow \frac{\pi}{4} \leq f(x) \leq \frac{3\pi}{4}$$

$$30. \sin \frac{\pi x_1}{2} = \sin \frac{\pi x_2}{2} \Rightarrow x_1 = x_2 \Rightarrow f \text{ is one-one}$$

$$31. f(2\pi + x) = f(x) \Rightarrow f \text{ is not one - one}$$

$$\text{Range of } f \neq \mathbb{C} \Rightarrow f \text{ is not onto}$$

$$32. f(1) = 0, f(2) = -1, f(3) = 1, f(4) = -2, \dots$$

$$\therefore f: \mathbb{N} \rightarrow \mathbb{Z} \text{ is one-one, onto}$$

$$33. f\left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right) = f\left(\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}\right) = 1$$

$$\Rightarrow f \text{ is not one-one}$$

$$\forall K \in \mathbb{R}, \text{ there exist a matrix } A = \begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix}$$

$$\text{such that } f(A) = K \Rightarrow f \text{ is onto}$$

$$34. \text{Given } f(x) = \frac{x-m}{x-n}$$

$$\text{where } m \neq n, \forall x \in \mathbb{R} - \{n\}, \text{ Let } x_1, x_2 \in \mathbb{R}$$

$$\therefore f(x_1) = f(x_2) \Rightarrow \frac{x_1 - m}{x_1 - n} = \frac{x_2 - m}{x_2 - n} \Rightarrow x_1 = x_2$$

$$\therefore f \text{ is one-one}$$

$$\text{Let } \lambda \in \mathbb{R} \text{ such that } f(x) = \lambda, \therefore \frac{x-m}{x-n} = \lambda$$

$$\therefore x = \frac{m - n\lambda}{1 - \lambda}$$

$$x \text{ is not defined for } \lambda = 1, \text{ also } x \text{ is not real.}$$

$$\therefore f(x) \text{ is not onto function.}$$

$$\text{If a function is not onto it referred that it is into function. Hence } f \text{ is one-one into function.}$$

$$35. (f-g)(x) = f(x) - g(x)$$

$$= \begin{cases} 0 - x = -x, & \text{if } x \text{ is rational} \\ x - 0 = x & \text{if } x \text{ is irrational} \end{cases}$$

$$\text{Clearly } (f-g)(x) \text{ is one-one and onto.}$$

$$f \circ g(x) = x^3 - \frac{1}{x^3},$$

$$f\left(x - \frac{1}{x}\right) = \left(x - \frac{1}{x}\right)^3 + 3\left(x - \frac{1}{x}\right)$$

$$\text{let } x - \frac{1}{x} = t, f(t) = t^3 + 3t$$

$$\text{Thus } f(x) = x^3 + 3x, f'(x) = 3x^2 + 3.$$

$$37. (f \circ g)(x) = f(g(x)) = a(cx + d) + b$$

$$\text{and } (g \circ f)(x) = g(f(x)) = c(ax + b) + d$$

$$(f \circ g)(x) = (g \circ f)(x) \text{ and at } a=1, b=2$$

$$\Rightarrow cx + d + 2 = cx + 2c + d \Rightarrow c = 1 \text{ and } d \text{ is arbitrary}$$

$$38. (g \circ f)(x) = |\sin x| \text{ and } f(x) = \sin^2 x$$

$$\Rightarrow g(\sin^2 x) = |\sin x| \therefore g(x) = \sqrt{x}$$

$$39. g(f(x)) \leq f(g(x)) \Rightarrow g(|x|) \leq f[x]$$

$$[|x|] \leq [x] \text{ this is true for } x \in \mathbb{R}.$$

$$40. \text{ Since } g(x) = 3 + 4x$$

$$\therefore g^2(x) = (g \circ g)(x) = g\{g(x)\} = g(3 + 4x)$$

$$= 3 + 4(3 + 4x) = 15 + 4^2 x = (4^2 - 1) + 4^2 x$$

$$g^3(x) = (g \circ g \circ g)x = g\{g^2(x)\} = g(15 + 4^2 x)$$

$$= 3 + 4(15 + 4^2 x) = 63 + 4^3 x = (4^3 - 1) + 4^3 x$$

$$\text{similarly, we get } g^n(x) = (4^n - 1) + 4^n x$$

$$41. \text{ Suppose } f(x)=1, \text{ then } f(y)=1, f(z)=2 \Rightarrow f \text{ is not an injection.}$$

$$\text{Suppose } f(y) \neq 1, \text{ then}$$

$$f(z) = 2, f(y) = 3, f(x) \neq 1 \text{ A contradiction}$$

$$\text{Suppose } f(z) \neq 2, \text{ then}$$

$$f(y)=1, f(z)=3, f(x)=2 \text{ this is true } \therefore f^{-1}(1) = y$$

$$42. y = 2^{x(y-1)} \Rightarrow x(x-1) = \log_2 y$$

$$\Rightarrow x^2 - x - \log_2 y = 0 \therefore x = \frac{1 \pm \sqrt{1 + 4 \log_2 y}}{2}$$

$$\therefore f^{-1}(x) = \frac{1}{2} \left\{ 1 + \sqrt{1 + 4 \log_2 y} \right\} \quad [\because x \geq 1]$$

$$43. \text{ Take } f(x) = 1 - x^n$$

$$44. \text{ Take } f(x) = x + 1 \text{ and verify}$$

$$45. f(x) = 2^x$$

$$46. f(x-y) = f(x)f(y) - f(a-x)f(a+y)$$

$$x=0, y=0 \Rightarrow f(0-0) = f(0)f(0) - f(a-0)f(a+0)$$

$$\Rightarrow f(0) = (f(0))^2 - (f(a))^2$$

$$\Rightarrow 1 = (1)^2 - (f(a))^2 \Rightarrow (f(a))^2 = 0 \Rightarrow f(a) = 0$$

$$\text{Now } f(2a-x) = f(a-(x-a))$$

$$= f(a)f(x-a) - f(a-a)f(a+x-a)$$



LEVEL - II (H.W)

REAL VALUED FUNCTIONS

$$1. \text{ If } f(x+ay, x-ay) = axy, \text{ then } f(x, y) =$$

$$1) xy \quad 2) x^2 - a^2 y^2 \quad 3) \frac{x^2 - y^2}{4} \quad 4) \frac{x^2 - y^2}{a^2}$$

$$2. \text{ If } f\left(2x + \frac{y}{8}, 2x - \frac{y}{8}\right) = xy, \text{ then}$$

$$f(m, n) + f(n, m) = 0$$

$$1) \text{ Only when } m = n \quad 2) \text{ Only when } m \neq n$$

$$3) \text{ Only when } m = -n \quad 4) \text{ For all } m \text{ and } n$$

EVEN AND ODD FUNCTIONS

$$3. \text{ Let } f(x) = |x-2| + |x-3| + |x-4| \text{ and}$$

$$g(x) = f(x+1). \text{ Then } g(x) \text{ is}$$

$$1) \text{ an even function} \quad 2) \text{ an odd function}$$

$$3) \text{ neither even nor odd} \quad 4) \text{ periodic}$$

$$4. \text{ If } f(x) = \text{sgn}\{x\} \text{ (where } \{.\} \text{ denotes the fractional part of } x), \text{ is}$$

$$1) \text{ even function} \quad 2) \text{ odd function}$$

$$3) \text{ neither even nor odd} \quad 4) \text{ constant function}$$

PERIODIC FUNCTIONS

$$5. \text{ period of}$$

$$f(x) = [x] + [2x] + [3x] + \dots + [nx] - \frac{n(n+1)}{2}x,$$

$$\text{where } n \in \mathbb{N} \text{ is}$$

$$1) n \quad 2) 1 \quad 3) \frac{1}{n} \quad 4) 2$$

$$6. \text{ Let } f(x) = \cos 3x + \sin \sqrt{3}x. \text{ Then } f(x) \text{ is}$$

$$1) \text{ a periodic function of period } 2\pi$$

$$2) \text{ a periodic function of period } \sqrt{3}\pi$$

$$3) \text{ not a periodic function}$$

$$4) \text{ a periodic function of period } \pi$$

7 The function $f(x) = x - [x] + \cos x$, where $[x]$ is the greatest integer less than or equal to x is a

- 1) periodic function of indeterminate period
- 2) periodic function of period 2π
- 3) nonperiodic function
- 4) periodic function of period 1

8. If $f(x) = e^{x - [x] + |\cos \pi x| + |\cos 2\pi x| + \dots + |\cos n\pi x|}$

then the period of $f(x)$ is

- 1) 1
- 2) $\frac{1}{n}$
- 3) $\frac{2}{n}$
- 4) no fundamental period

9. If $f(x)$ and $g(x)$ are periodic functions with period 7 and 11, respectively. Then the period

of $F(x) = f(x)g\left(\frac{x}{5}\right) - g(x)f\left(\frac{x}{3}\right)$ is

- 1) 177
- 2) 222
- 3) 433
- 4) 1155

DOMAIN OF THE FUNCTION

10. The domain of $f(x) = \frac{1}{\sqrt{9-x^2}} + \sqrt{x^2-4}$ is

- 1) $(-4, -2) \cup (2, 4)$
- 2) $(-3, -2] \cup [2, 3)$
- 3) $(-\infty, -3) \cup (2, \infty)$
- 4) $(-\infty, \infty)$

11. Domain of $\frac{1}{\sqrt{[x]^2 - [x] - 2}}$

- 1) $\mathbb{R} / [-1, 3)$
- 2) $(-\infty, -3) \cap [3, \infty)$
- 3) $[2, \infty)$
- 4) $(-\infty, 3]$

12. Domain of $\sqrt{\log_{10}\left(\frac{3-x}{x}\right)}$

- 1) $\left(0, \frac{3}{2}\right]$
- 2) $\left(-\infty, \frac{3}{2}\right]$
- 3) $(0, 3)$
- 4) $\mathbb{R}$

13. Domain of $f(x) = \log(x - [x])$

- 1) $\mathbb{R}$
- 2) $\mathbb{Z}$
- 3) $\mathbb{R} - \mathbb{Z}$
- 4) $(0, \infty)$

14. The domain of $\sqrt{\log_{\frac{x}{e}} + 1}$ is

- 1) $\left(0, \frac{3}{e}\right]$
- 2) $\left(-\infty, \frac{3}{e}\right]$
- 3) $\left[\frac{3}{e}, \infty\right)$
- 4) $\mathbb{R}$

15. The domain of the function

$$f(x) = \sqrt{\frac{x-2}{x+2}} + \sqrt{\frac{1-x}{1+x}}$$

- 1) $\mathbb{R}$
- 2) $[-2, 2]$
- 3) $[-1, 1]$
- 4) $\emptyset$

16. The domain of $f(x) = \frac{1}{1 - 2 \sin x}$ is

$$1) (-\infty, \infty) \quad 2) \mathbb{R} - \left\{n\pi + (-1)^n \frac{\pi}{6} : n \in \mathbb{Z}\right\}$$

$$3) \mathbb{R} - \{n\pi : n \in \mathbb{Z}\}$$

$$4) \mathbb{R} - \left\{n\pi + (-1)^n \frac{\pi}{3} : n \in \mathbb{Z}\right\}$$

17. $f(x) = \begin{cases} [x], & -3 < x \leq -1 \\ |x|, & -1 < x < 1 \\ [x], & 1 \leq x < 3 \end{cases}$ then

$$\{x : f(x) \geq 0\} = \quad \text{(EAM-04)}$$

$$1) (-1, 3) \quad 2) [-1, 3) \quad 3) (-1, 3] \quad 4) [-1, 3]$$

18. The domain of $f(x) = \sqrt{2 - \log_3(x-1)}$ is

$$1) (2, 12] \quad 2) (-\infty, 10] \quad 3) (3, 12] \quad 4) (1, 10]$$

19. The domain of $f(x) = \frac{\sqrt{2+x} + \sqrt{2-x}}{x}$ is

$$1) (-2, 2) \quad 2) [-2, 0) \cup (0, 2]$$

$$3) [-2, 2] \quad 4) (-\infty, 2)$$

20. Let f and g be two real functions given by

$$f = \{(0, 1), (2, 0), (3, -4), (4, 2), (5, 1)\}$$

$$g = \{(1, 0), (2, 2), (3, -1), (4, 4), (5, 3)\}$$

then the domain of $f \cdot g$ is given by

$$1) \{1, 2, 3, 4\} \quad 2) \{1, 2, 3, 4, 5\}$$

$$3) \{0, 1, 2, 3, 4\} \quad 4) \{2, 3, 4, 5\}$$

21. The domain of the function f defined by

$$f(x) = \sqrt{4-x} + \frac{1}{\sqrt{x^2-1}}$$

$$1) (-\infty, -1) \cup (1, 4] \quad 2) (-\infty, -1] \cup (1, 4]$$

$$3) (-\infty, -1] \cup [1, 4] \quad 4) (-\infty, -1) \cup [1, 4)$$

22. Domain of $\sqrt{\frac{4-x^2}{[x]+2}}$

$$1) -\infty, -2) \cup (2, \infty) \quad 2) \mathbb{R} - \{-2\}$$

$$3) [-1, 2) \quad 4) (-\infty, -2) \cup [-1, 2]$$

23. The set of all real numbers satisfying

$$e^{\left(\frac{1}{x}-1\right)} < 1 \text{ is}$$

$$1) (0, \infty) \quad 2) (-\infty, 0) \cup (1, \infty) \quad 3) (-\infty, \infty) \quad 4) (0, 1)$$

RANGE OF THE FUNCTION

24. The range of $f(x) = 8\sqrt{2} \sin \sqrt{\frac{\pi^2}{16} - x^2}$ is
1) $[-1, 1]$ 2) $[0, 1]$ 3) $[0, 8]$ 4) $[0, 4]$
25. The range of $f(x) = \frac{x^4}{1+x^8}$ is
1) $[0, \infty)$ 2) $\left[0, \frac{1}{2}\right]$ 3) $[0, 1]$ 4) $(-\infty, \infty)$
26. The range of $f(x) = \frac{1 - \tan x}{1 + \tan x}$ is
1) $(-\infty, \infty)$ 2) $(-\infty, 0)$
3) $(0, \infty)$ 4) $(-\infty, -1) \cup (-1, \infty)$
27. If $a^2 + b^2 + c^2 = 1$ then the range of $ab + bc + ca$ is
1) $[1, \infty)$ 2) $\left[-\frac{1}{2}, \infty\right]$ 3) $\left(-\frac{1}{2}, 1\right)$ 4) $\left[-\frac{1}{2}, 1\right]$
28. The range of the function $f(x) = {}^{7-x}P_{x-3}$ is
[AIE-2004]
1) $\{1, 2, 3\}$ 2) $\{1, 2, 3, 4, 5, 6\}$
3) $\{1, 2, 3, 4\}$ 4) $\{1, 2, 3, 4, 5\}$
29. Range of $f(x) = \sin^{-1} x + \sec^{-1} x$ is
1. $[-\pi/2, \pi/2]$ 2. $[0, \pi] - \{\pi/2\}$
3. $\{\pi/2\}$ 4. $\{\pi/1\}$

TYPES OF FUNCTIONS

30. If $f(x) = |x-1| + |x-2| + |x-3|$ when $2 < x < 3$ is
1) one one function only 2) an onto function only
3) into function 4) identify function
31. $f: (0, \infty) \rightarrow (0, \infty)$ defined by
 $f(x) = \begin{cases} 2^x, & x \in (0, 1) \\ 5^x, & x \in [1, \infty) \end{cases}$ is
1) one-one but not onto 2) onto but not one-one
3) neither one - one nor onto 4) bijective
32. The function $f: (-\infty, \infty) \rightarrow (-\infty, \infty)$ defined by
 $f(x) = e^{|x|}$ is
1) one-one but not onto 2) onto but not one-one
3) neither one - one nor onto 4) bijective
33. $f: C \rightarrow C$ is defined as $f(x) = \frac{ax+b}{cx+d}$,
 $bd \neq 0$ then f is a constant function when
1) $a = c$ 2) $b = d$ 3) $ad = bc$ 4) $ab = cd$

34. $f: R \rightarrow R, f(x) = x|x|$ is
1) one-one but not onto 2) one-one onto
3) onto but not one-one 4) neither one-one nor onto
35. Let $f: \{x, y, z\} \rightarrow [a, b, c]$ be a one-one function and only one of the conditions (i) $f(x) \neq b$, (ii) $f(y) = b$, (iii) $f(z) \neq a$ is true then the function f is given by the set
1) $\{(x, a), (y, b), (z, c)\}$ 2) $\{(x, a), (y, c), (z, b)\}$
3) $\{(x, b), (y, a), (z, c)\}$ 4) $\{(x, c), (y, b), (z, a)\}$
36. $f: N \rightarrow N$ where $f(x) = x - (-1)^x$ then f is
1) one-one and into 2) many-one and into
3) one-one and onto 4) many-one and onto

COMPOSITE FUNCTION

37. If $f(x) = \log_a x$ and $F(x) = a^x$, then
 $F[f(x)]$ is
1) $f[F(x)]$ 2) $f[F(2x)]$
3) $f|F(2x)|$ 4) $F(x)$
38. Suppose that $g(x) = 1 + \sqrt{x}$ and
 $f(g(x)) = 3 + 2\sqrt{x} + x$, then $f(x)$ is
1) $1 + 2x^2$ 2) $2 + x^2$ 3) $1 + x$ 4) $2 + x$
39. Let $g(x) = 1 + x - [x]$ and
 $f(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0, \\ 1, & x > 0 \end{cases}$ then for all $x, f(g(x)) =$
1) x 2) 1 3) $f(x)$ 4) $g(x)$
40. If $\square$ and $\square$ are defined by
 $\square$ and $\square$ then the
values of x such that $\square$ are.
1) $1, 2$ 2) $-1, 2$ 3) $-1, -2$ 4) $1, -2$
- INVERTIBILITY & INVERSE OF A FUNCTIONS
41. If $f(x) = 1 + x + x^2 + x^3 + \dots$ for $\square$ then $\square$
1) $\square$ 2) $\square$ 3) $\square$ 4) $\square$

42. If then domain of for the existence of inverse of (EAM-88)

- 1) $[0, \infty)$ 2) 3) 4)

43. Let Then is invertible for

- 1) 2) 3) 4)

FUNCTIONAL TRANSFORMATIONS

44. If then

- 1) -2 2) -1 3) $1/2$ 4) 0

45. If and then

- 1) 35 2) 36 3) 4) 128

46. If $f(x)$ is a function such that

$f(xy) = f(x) + f(y)$ and $f(2) = 1$ then

- 1) x^2 2) 2^x 3) $\log_2 x$ 4) $\log_x 2$

47. If f satisfies the relation

- and
1) 0 2) 1 3) 2 4) 3

LEVEL-II(H.W.)-KEY

- | | | | | | |
|-------|-------|-------|-------|-------|-------|
| 01) 3 | 02) 4 | 03) 3 | 04) 1 | 05) 2 | 06) 3 |
| 07) 3 | 08) 1 | 09) 4 | 10) 2 | 11) 1 | 12) 1 |
| 13) 3 | 14) 1 | 15) 4 | 16) 2 | 17) 1 | 18) 4 |
| 19) 2 | 20) 4 | 21) 1 | 22) 4 | 23) 2 | 24) 3 |
| 25) 2 | 26) 4 | 27) 4 | 28) 1 | 29) 3 | 30) 4 |
| 31) 1 | 32) 3 | 33) 3 | 34) 2 | 35) 3 | 36) 3 |
| 37) 1 | 38) 2 | 39) 2 | 40) 3 | 41) 4 | 42) 2 |
| 43) 3 | 44) 4 | 45) 4 | 46) 3 | 47) 1 | |

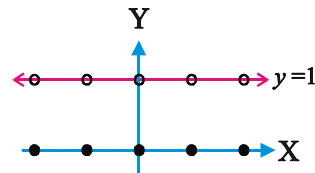
LEVEL-II(H.W.)-HINTS

1. Given
Let and
Then
Substituting the value of x and y in (i), We obtain

2. Let and , then and Given

for all m, n

3.
If
If
If
If



4.

5.

period of

6. has the period and has the period .

As and do not have a common multiple, is not periodic

7. has the period 1 and has the period . Clearly, 1 and do not have a common multiple.

8. Period of
 period of is half of that of
L.C.M of is 1].

FUNCTIONS

10. $9 - x^2 > 0$ and

11.

13.

14. and $x > 0$

16. 

18. $f(x) = [x] > 0$ for and $x - 1 > 0$

19.

20. Common domain of f and g .

21. f

22.

--

24.

JEE-MAIN-JR-MATHS VOL-I

(uses (ii) also)

(uses (i) also)

(uses (i) also)

(uses (i) also)

uses (i) also)

possible)

(uses (i) also)

36. , when is clearly are
one-one and onto

37.

38.

Put

Then

therefore,

39. Here

(where)

Now

Clearly

.so,

40.

41. Let

$$f^{-1}(y) = x$$

42.

is a bijection in

43. By defination of composition of functions

We know is bijection only, when

Thus is bijection if

44. The given ex;pression is

Apply

45. Take $f(x) = k^x$

46. Take $f(x) = \log_a x$

47. If $y = x$ then
if $y = -x$ then



LEVEL - III

1. If for nonzero x ,

where , then $f(2) =$

1)

2)

3)

4)

2. If and a, b are the roots of

, then

1)

2)

3)

4)

3. If $\square$ then $\square =$
 1) 997 2) 997.5 3) 998 4) 998.5
4. If $[x]$ stands for the greatest integer function, then $\square$
 1) 498 2) 499 3) 500 4) 501
5. Let $\square \in \mathbb{R}$ where $\square$ be an odd function then the value of a is (where $[.]$ represents greatest integer function)
 1) less than 11 2) 11 3) greater than 11 4) 12
6. If $\square$ for all real $\square$ and $\square$, then the function $\square$ is
 1) even function 2) odd function
 3) odd if $\square$ 4) neither even nor odd
7. Let $\square$, then $\square$ is (where $[.]$ represents greatest integer function)
 1) an odd function 2) an even function
 3) both odd and even 4) neither odd nor even
8. If the real valued function $\square$ is even, then $n =$
 1) 2 2) $\square$ 3) $\square$ 4) 3
9. If f is an even function defined on the interval $\square$ then find the total number of real values of x satisfying the equations $\square$ are
 1) 1 2) 2 3) 4 4) 8

10. Let the function $\square$ be defined on the interval $[0, 1]$. The even extension of $\square$ of the interval $[-1, 1]$ is
 1) $\square$
 2) $\square$
 3) $\square$
 4) $\square$
11. If $\square$ is an invertible function such that $\square$ and $\square$ are symmetric about the line $\square$ then
 1) $\square$ is odd
 2) $\square$ and $\square$ may not be symmetric about the line $\square$
 3) $\square$ may not be odd 4) $\square$ may be odd
12. If $\square$ is a function satisfying the property $\square$, then the period of $\square$ is
 1) 2 2) 4 3) 8 4) 12
13. If the period of the function $\square$ where $[n]$ denotes the greatest integer less than or equal to n is $\square$, then
 1) $\square$ 2) $\square$
 3) $\square$ 4) $\square$
14. The period of the function $\square$ is
 1) 1 2) 2 3) Non periodic 4) 4
15. The period of the function $\square$ is
 1) $\square$ 2) $\square$ 3) $\square$ 4) $\square$
16. The period of the function $\square$ is
 1) $\square$ 2) $\square$ 3) $\square$ 4) $\square$

17. The period of the function

_____ is

1) _____ 2) _____ 3) _____ 4) _____

18. If $f(x)$ is an odd periodic function with period 2, then $f(4) =$

- 1) _____ 2) 4 3) 2 4) 0

19. If f is periodic g is polynomial function and

_____ is periodic and _____

then _____ is

1) 13 2) 15 3) 11 4) 16

20. The domain of the function _____ is

- 1) _____ 2) _____ 3) _____ 4) $\mathbb{R}$

21. The largest interval lying in _____ for which the function is

_____ defined, is

[AIE-2007]

- 1) _____ 2) _____ 3) _____ 4) _____

22. The domain of _____ is

- 1) _____ 2) _____ 3) _____ 4) _____

23. The domain of

_____ is

1) _____ 2) _____

3) _____ 4) _____

24. The domain of _____ is

- 1) _____ 2) _____
- 3) _____ 4) _____

25. If _____ and _____, then domain of _____ is

- 1) _____
- 2) _____
- 3) _____
- 4) _____

26. Let _____ and _____, the domain of _____ is

- 1) _____ 2) _____ 3) _____ 4) $\mathbb{R}$

27. The function

_____ is defined on the set S , then S is equal to

1) _____ 2) _____ 3) _____ 4) _____

28. If _____ then the domain of _____ is

- 1) _____ 2) _____
- 3) _____ 4) _____

29. If _____ is defined on _____, then the domain of _____ is

- 1) _____
- 2) _____
- 3) _____
- 4) _____

30. Domain of

- 1) _____ 2) _____
- 3) _____ 4) _____

31. The domain of _____ is
- 1) _____ 2) _____
- 3) _____ 4) _____
32. If _____ and _____, then P lies interval
- 1) _____ 2) _____ 3) _____ 4) _____
33. The range of _____ is
- 1) _____ 2) _____
- 3) _____ 4) _____
34. The image of the interval _____ under the mapping _____, given by _____ is
- 1) _____ 2) _____ 3) _____ 4) _____
35. The range of _____ is
- 1) _____ 2) _____ 3) _____ 4) _____
36. If _____, then range of _____ is
- 1) _____ 2) _____ 3) _____ 4) _____
37. The range of _____ is
- 1) _____ 2) _____ 3) _____ 4) R
38. The range of _____ is
- 1) _____ 2) _____ 3) _____ 4) _____

39. If _____ is defined by _____ for _____, where _____ is the greatest integer not exceeding _____, then the range of _____ is
- 1) _____ 2) _____
- 3) _____ 4) _____
40. If _____ are real constants) and _____ then the range of _____ is
- 1) _____ 2) _____ 3) _____ 4) _____
41. The range of _____ where _____ denotes the greatest integer function, is
- 1) _____ 2) _____ 3) _____ 4) _____
42. The range of _____, where _____ denotes the greatest integer function _____ is
- 1) _____ 2) _____ 3) _____ 4) _____
43. The function _____ is defined by _____ where [.] is G.I.F is surjective then B =
- 1) R 2) _____ 3) _____ 4) _____
44. If _____, and _____ then the mapping _____ is
- 1) One - one but not onto
- 2) Onto but not one - one
- 3) Both one - one and onto
- 4) Neither one - one nor onto
45. The functions _____ is defined by _____ is
- 1) One one onto 2) One one into
- 3) Many one onto 4) Many one into

46. Let S be the set of all triangles and $\mathbb{R}^+$ be the set of positive real numbers. Then the function

$f: S \rightarrow \mathbb{R}^+$ where $f(T) = \frac{\text{Area}(T)}{\text{Perimeter}(T)}$ is

- 1) injective but not surjective
- 2) surjective but not injective
- 3) injective as well as surjective
- 4) neither injective nor surjective

47. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ is defined by

$f(x) = \begin{cases} x & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$ then f is

- 1) one-one onto
- 2) one-one into
- 3) many one onto
- 4) many one into

48. If $f: \mathbb{R} \rightarrow \mathbb{R}$ then $f(x) = \begin{cases} x & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$

- 1) f is one-one and onto
- 2) f is one-one but not onto
- 3) f is onto but not one-one
- 4) f is neither one-one nor onto

49. If $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ then $(f \circ g)(x) = f(g(x))$

- 1) 1
- 2) 0
- 3) $\sin x$
- 4) $\cos x$

50. If $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ and

$(f \circ g)(x) = f(g(x))$ then which is $(g \circ f)(x)$

- 1) $f(g(x))$
- 2) $g(f(x))$
- 3) $f(f(x))$
- 4) $g(g(x))$

51. If $f: \mathbb{R} \rightarrow \mathbb{R}$ is a function defined by

$f(x) = \begin{cases} x & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$ where $[.]$ is G.I.F then

- 1) $f(f(x)) = f(x)$
- 2) $f(f(x)) = x$
- 3) $f(f(x)) = 0$
- 4) $f(f(x)) = x^2$

52. If the function $f: \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = \begin{cases} x & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$ then $(f \circ f)(x) =$

- 1) x
- 2) 0
- 3) x^2
- 4) x^3

53. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ then the solution of the equation $f(x) = 0$ is

- 1) $x = 0$
- 2) $x = 1$
- 3) $x = -1$
- 4) $x = 0$

54. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be given by

$f(x) = \begin{cases} x & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$ Then $f(f(x)) =$

- 1) x
- 2) 0
- 3) does not exist because f is not one-one
- 4) does not exist because f is not onto

55. The inverse of the function

$f: \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = \begin{cases} x & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$

- 1) $f^{-1}(x) = x$
- 2) $f^{-1}(x) = 0$
- 3) $f^{-1}(x) = \begin{cases} x & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$
- 4) $f^{-1}(x) = \begin{cases} x & \text{if } x \geq 0 \\ x & \text{if } x < 0 \end{cases}$

56. If $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ and

$(f \circ g)(x) = f(g(x))$ then $(g \circ f)(x) =$

- 1) a
- 2) b
- 3) ab
- 4) $3ab$

LEVEL-III-KEY

- | | | | | | |
|-------|-------|-------|-------|-------|-------|
| 01) 2 | 02) 3 | 03) 2 | 04) 3 | 05) 3 | 06) 1 |
| 07) 1 | 08) 4 | 09) 3 | 10) 1 | 11) 1 | 12) 2 |
| 13) 1 | 14) 3 | 15) 3 | 16) 1 | 17) 2 | 18) 4 |
| 19) 3 | 20) 1 | 21) 4 | 22) 2 | 23) 3 | 24) 2 |
| 25) 4 | 26) 3 | 27) 1 | 28) 3 | 29) 1 | 30) 3 |
| 31) 4 | 32) 1 | 33) 2 | 34) 2 | 35) 1 | 36) 2 |
| 37) 1 | 38) 2 | 39) 2 | 40) 1 | 41) 2 | 42) 3 |
| 43) 4 | 44) 3 | 45) 4 | 46) 2 | 47) 1 | 48) 1 |
| 49) 1 | 50) 3 | 51) 2 | 52) 2 | 53) 1 | 54) 1 |
| 55) 4 | 56) 3 | | | | |

LEVEL-III-HINTS

1. Replace x by $\frac{1}{x}$ then solve.

2. We have,

$\frac{1}{x} = \frac{1}{f(x)}$

[since a, b all roots of $x^2 - 1 = 0$, $\therefore \frac{1}{a} = a$, $\frac{1}{b} = b$]

Similarly

3.

4. if

5.

Now given,

6. Given . Put , then . Put , then

Now,

7. The denominator is

8.

9.

10.

11. Since and are symmetric about the line

If lies on then lie on
lies on is odd.

12. We have

Replance x by x + 1,

Replace x by x + 2

From (1) and (3)

We get

i.e.,

is periodic with period 2k.

13. Sin x is a periodic function with period , therefore is a periodic function

with period

But the period of f(x) is (given)

14. Let $y = \sin nx$.
Then, $\frac{dy}{dx} = n \cos nx$ where n changes with x clearly
no constant $k > 0$ is possible for which
 $\frac{dy}{dx} = k$ corresponding to all x .
 $y = \sin nx$ is a non periodic function.
15. We have $y = \sin nx$
 $\frac{dy}{dx} = n \cos nx$
for all x
 $y = \sin nx$ is periodic with period $\frac{2\pi}{n}$.
16. Since $\sin x$ and $\cos x$ are periodic function with
period 2π .
 $\sin x$ and $\cos x$ are periodic functions with
period 2π and period of $\sin x$ is 2π
of $\cos x$ is 2π .
17. Since $y = \sin nx$ is a periodic function with
period $\frac{2\pi}{n}$,
 $y = \sin nx$ is a periodic function
with period $\frac{2\pi}{n}$.
18. $y = \sin nx$; $y = \cos nx$
19. From the given data $y = \sin nx$ must be linear function.
Hence, $y = \sin nx$
Also $y = \cos nx$ and $y = \sin nx$
Solving, we get $y = \sin nx$ and $y = \cos nx$
Hence, $y = \sin nx$ Then $y = \cos nx$.
20. $y = \sin nx$ is defined for $a, b > 0$ and $\frac{2\pi}{n}$
21. $y = \sin nx$ is defined if $y = \sin nx$
 $y = \sin nx$ is defined if $y = \sin nx$
 $y = \sin nx$ is defined if $y = \sin nx$

- From (1) and (2): $y = \sin nx$
22. $y = \sin nx$ is defined if $y = \sin nx$
- Domain of $y = \sin nx$
23. $y = \sin nx$ is defined then $y = \sin nx$
24. $y = \sin nx$ is defined if $y = \sin nx$
25. We have,
 $y = \sin nx$
 $y = \sin nx$
 $y = \sin nx$
26. We have domain of f is $y = \sin nx$
For the function $y = \sin nx$
 $y = \sin nx$
 $y = \sin nx$
domain of $y = \sin nx$ is $y = \sin nx$
domain of $y = \sin nx$ is $y = \sin nx$
27. For the two components to be meaningful, we
must have $y = \sin nx$
Hence, $y = \sin nx$
 $y = \sin nx$
28. $y = \sin nx$
 $y = \log y$
since $a > 0$ $y = \sin nx$ is defined if $y = \sin nx$ and
 $y = \sin nx$

29. Since the domain of f is $(0,1]$

30. _____
31. For _____ to be defined, we must have

 Also, _____
 Now, _____

 Also _____
 Thus, _____
32. We have, _____

 Also, _____

33. _____ is defined if _____ : _____

 _____ Range of _____
34. Since the given function has minimum value 75 which is attained at $x = 2$ and maximum value is 89 which is attained at $x = 3$ Hence the range of f is $[75, 89]$
35. If $x > 0$ then $f(x) = 0$, If $x < 0$ then $f(x) = \tan hx$
 If _____ then _____
 _____ Range does not contain -1
 _____ Range = $(-1,0]$

36. When _____ is rational say _____, then _____ is a multiple of _____ and _____

 When _____ is irrational, _____ multiple of _____
 _____,

 Thus _____ when _____ is irrational
 =1, when _____ is rational.
 _____ Range _____
37. _____
38. _____, _____
39. _____ or
 _____ or
 _____ or

40. _____ which has
 the range _____
41. Thus, from domain point of view,

 or _____.
42. _____ where

 _____ (as p is an integer)

Now, for

The range of is 1.

43. verify by taking integer & decimal number.

44. Let , then

☐ which is bijective in

Hence is bijective

45. Clearly from the graph, f is many one into function

46. Two triangles may have equal areas

is not one-one.

Since each positive real number can represent area of triangle. is onto.

47.

 since -1 ,
 let
 since and

is an increasing function therefore said to be 1-1 function, Now and

is continuous also

☐ Range of codomain of

48.

(impossible)

49.

50.

or

51.

52. Let $f(x) = y$

53.

54. Let $y = f(x)$

55. Let $y = f(x)$

56. Put . Then

LEVEL - IV

Assertion & Reason Type Q - 1 to Q - 5

- a) Assertion is True, Reason is True, Reason is a correct explanation for Assertion
b) Assertion is True, Reason is True, Reason is not a correct explanation for Assertion
c) Assertion is True, Reason is False.
d) Assertion is False, Reason is True.

1. **Assertion :** The domain of

_____ , is _____

Reason : The domain of _____ is (0, 4).

- 1) a 2) b 3) c 4) d

2. **Assertion :** Every function can be uniquely expressed as the sum of an even function and an odd function.

Reason : The set of values of parameter 'a' for which the function $f(x)$ defined as

_____ on the set $[-3, 3]$ is an odd function is, _____.

- 1) a 2) b 3) c 4) d

3. **Assertion :** The domain of the function

_____ is _____.

Reason : If _____, then _____

cannot attain the value _____.

- 1) a 2) b 3) c 4) d

4. **Assertion :** The function $f(x)$ given by

_____ is an odd function.

Reason : The composite function of two odd functions is an odd function

- 1) a 2) b 3) c 4) d

5. **Assertion :** If $\{x\}$ and $[x]$ represent fractional part and integral part of x then

_____ .

Reason : _____ and _____, where I is integer

- 1) a 2) b 3) c 4) d

STATEMENTS

- 1) only I is true 2) only II is true
3) both I and II are true
4) neither I nor II true

6. **Statement I:** The product of two odd functions is an even function.

Statement II: A constant function is always a bijection.

7. **Statement I :** Every strictly monotonic function is one one

Statement II : The function _____ defined by _____ is one one.

8. Let _____ where $[.]$ denotes the greatest integer function

I : Domain of $f(x)$ is $\mathbb{R}$

II : Range of $f(x)$ is _____

9. **Statement I :** The range of the function

_____ is $\{0\}$

Statement II : The range of the function

_____ is _____.

10. If Q denotes the set of all rational numbers

and _____ for any _____, then observe the following statements

I. _____ is real for each _____

II. _____ is a complex number for each _____

Which of the following is correct ?

- 1) Both I and II are true 2) I is true, II is false
3) I is false, II is true 4) Both I and II are false

MATCHINGS

11. If $f(x) = \frac{x^2 - 1}{x^2 + 1}$ is defined by

then the

correct matching of List I from List II is

List - I

List - II

- A) $f(x) = \frac{x^2 - 1}{x^2 + 1}$ i) 14
 B) $f(x) = \frac{x^2 + 1}{x^2 - 1}$ ii) 4
 C) $f(x) = \frac{x^2 - 1}{x^2 + 1}$ iii) -11
 D) $f(x) = \frac{x^2 + 1}{x^2 - 1}$ iv) -1
 v) 1
 vi) 0

- 1) A - iii; B - vi; C - ii; D - v
 2) A - iii; B - iv; C - ii; D - v
 3) A - iv; B - iii; C - ii; D - i
 4) A - iii; B - vi; C - v; D - ii

12. Match the following

List - I

List - II

- I. $f(x) = \frac{x^2 - 1}{x^2 + 1}$ defined by $f(x) = \frac{x^2 - 1}{x^2 + 1}$ is a) injection but not surjection
 II. $f(x) = \frac{x^2 + 1}{x^2 - 1}$ defined by $f(x) = \frac{x^2 + 1}{x^2 - 1}$ is b) surjection but not injection
 III. $f(x) = \frac{x^2 - 1}{x^2 + 1}$ defined by $f(x) = \frac{x^2 - 1}{x^2 + 1}$ is c) bijection
 IV. $f(x) = \frac{x^2 + 1}{x^2 - 1}$ defined by $f(x) = \frac{x^2 + 1}{x^2 - 1}$ is d) neither injection nor surjection
 1) c, d, a, b 2) c, d, b, a
 3) d, c, b, a 4) d, c, a, b

13. If $f(x) = \frac{x^2 - 1}{x^2 + 1}$ are functions from $\mathbb{R}$ to $\mathbb{R}$ then

List - I

List - II

- I. $f(x) = \frac{x^2 - 1}{x^2 + 1}$ a) even function
 II. $f(x) = \frac{x^2 + 1}{x^2 - 1}$ b) odd function
 III. $f(x) = \frac{x^2 - 1}{x^2 + 1}$ c) both even and odd function
 IV. $f(x) = \frac{x^2 + 1}{x^2 - 1}$ d) neither even nor odd function
 1) I-b; II-a; III-c; IV-d 2) I-a; II-b; III-d; IV-c
 3) I-a; II-b; III-c; IV-d 4) I-b; II-a; III-d; IV-c

14. let $\mathbb{R}$ denote the set of all real numbers and $\mathbb{R}^+$ denote the set of all positive real numbers. For the subsets A and B of $\mathbb{R}$ define $f: A \rightarrow B$ by $f(x) = \frac{x^2 - 1}{x^2 + 1}$ for $x \in A$. Observe the two lists given below.

List - I

List - II

I. f is one-one and

onto if

a) $A = \mathbb{R}$

II. f is one-one but not

onto if

b) $A = B = \mathbb{R}$

III. f is onto but not

one-one if

c) $A = \mathbb{R}, B = \mathbb{R}^+$

IV. f is neither one-one

nor onto if

d) $A = B = \mathbb{R}^+$

The correct matching of List I to List II is

- 1) I-a; II-b; III-c; IV-d 2) I-d; II-b; III-a; IV-c
 3) I-d; II-a; III-c; IV-b 4) I-d; II-b; III-c; IV-a

15. Observer the following lists

List - I

List - II

Functions

Domain

I. $f(x) = \frac{x^2 - 1}{x^2 + 1}$

a) $\mathbb{R}$

II. $f(x) = \frac{x^2 + 1}{x^2 - 1}$

b) $\{1\}$

III. $f(x) = \frac{x^2 - 1}{x^2 + 1}$

c) $\{-1\}$

IV. $f(x) = g(x) + h(x)$

d) $[1, 2]$

e) $\mathbb{R}$

The correct matching of List I to List II is

- 1) I-b; II-a; III-c; IV-d 2) I-a; II-b; III-d; IV-c
 3) I-d; II-e; III-a; IV-a 4) I-b; II-a; III-d; IV-c

LEVEL-IV-KEY

- 01) 3 02) 4 03) 1 04) 1 05) 1 06) 1
 07) 3 08) 3 09) 3 10) 3 11) 1 12) 2
 13) 4 14) 3 15) 3

LEVEL-IV-HINTS

1. $f(x) = \frac{x^2 - 1}{x^2 + 1}$ is defined for $x \in \mathbb{R}$
 2. $f(-x) = -f(x)$
 3. Denominator $x^2 + 1$
 4. $f(-x) = -f(x)$
 5. $f(x) = \frac{x^2 - 1}{x^2 + 1}$, increasing on $(0, 1)$
 6. $f(x) = \frac{x^2 + 1}{x^2 - 1}$
 7. Choose numerical values, verify
 8. Substitute values
 9. $f(x) = \frac{x^2 - 1}{x^2 + 1}$ is defined for $f(x)$ has to be defined and